

Comparative transcriptomics reveal a novel tardigrade specific DNA binding protein induced in response to ionizing radiation

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Abstract

Tardigrades, microscopic animals found in virtually all ecosystems, are renowned for their remarkable ability to withstand extreme conditions. Recent studies have identified novel tardigrade specific protein families that aid in resistance to desiccation and ionizing radiation (IR). Notably, a tardigrade specific DNA binding protein called Dsup (for DNA damage suppressor) has been found to protect from X-ray damage in human cells and from hydroxyl radicals *in vitro*. However, Dsup has only been found in two species within the Hypsibiodea superfamily.

To better understand mechanisms underlying radio-resistance in the Tardigrada phylum, we first characterized DNA damage and repair in response to IR in the model species *Hypsibius exemplaris*. By analysis of phosphorylated H2AX, we demonstrated the induction and repair of DNA double-strand breaks after IR exposure. Importantly, the rate of single-strand breaks induced was roughly equivalent to that in human cells, suggesting that DNA repair plays a predominant role in the remarkable radio-resistance of tardigrades. In order to identify novel tardigrade specific genes involved, we next conducted a comparative transcriptomics across three species, *H. exemplaris*, *Acutuncus antarcticus* and *Paramacrobiotus fairbanksi*, the latter belonging to the Macrobiotodea superfamily known to lack Dsup homologs. In all three species, many genes of DNA repair were among the most strongly overexpressed genes alongside a novel tardigrade specific gene, named Tardigrade DNA damage Response protein 1 (TDR1). We found that TDR1 protein interacts with DNA and forms aggregates at high concentration suggesting it may condensate DNA and act by preserving chromosome organization until DNA repair is accomplished. Remarkably, when expressed in human cells, TDR1 improved resistance to Bleomycin, a radiomimetic drug. Based on these findings, we

propose that TDR1 is a novel tardigrade specific gene responsible for conferring resistance to IR. Our study sheds light on mechanisms of DNA repair helping to cope with high levels of DNA damage. Furthermore, it suggests that at least two tardigrade specific genes, respectively for Dsup and TDR1, have independently evolved DNA-binding functions that contribute to radio-resistance in the Tardigrada phylum.

eLife assessment

This study offers **valuable** insight into the remarkable resistance of tardigrades to ionizing radiation by showing that radiation treatment induces a suite of DNA repair proteins and by identifying a strongly induced tardigrade-specific DNA-binding protein that can reduce the number of double-strand breaks in human U2OS cells. The evidence of upregulation of repair proteins is **convincing**, and the case for a role of the newly identified protein in repair can be strengthened as genetic tools for tardigrades become better developed. The results will interest the fields of DNA repair and radiobiology as well as tardigrade biologists.

Introduction

Tardigrades are microscopic animals found in marine or freshwater environments, and in semi-terrestrial habitats such as moss, lichen and leaf litter. They are well known for their resistance to IR (Jönsson 2019 [↗](#)) and extreme conditions like desiccation, freezing, and osmotic stress (Guidetti, Altiero, and Rebecchi 2011 [↗](#)). With over 1400 species, they belong to the clade of ecdysozoans, which also includes nematodes and arthropods (Degma and Guidetti, 2023 [↗](#)). Tardigrades share a highly conserved body plan, with a soft body protected by a cuticle, four pairs legs and a characteristic feeding apparatus. Tardigrades, however, can differ in their resistance to extreme conditions. For example, *Ramazzottius oberhaeuseri* withstands extremely rapid desiccation while *Hypsibius dujardini* only survives gradual dehydration (Wright 1989 [↗](#)), and the freshwater *Thulinus ruffoi* is not resistant to desiccation (Kondo et al. 2020 [↗](#)). Many species across the Tardigrada phylum can tolerate irradiation doses higher than 4000 Gy (Hashimoto and Kunieda 2017 [↗](#)), but some are less tolerant, like *Echiniscoides sigismundi*, which has an LD50 at 48 hours of 1000 Gy (Jönsson et al. 2016 [↗](#)). However, the doses compatible with maintenance of fertility seem much lower, e.g. the maximum is 100 Gy for *H. exemplaris* (Beltrán-Pardo et al. 2015 [↗](#)). Due to challenges in rearing tardigrades in the laboratory (Altiero and Rebecchi 2001 [↗](#)), the maintenance of fertility has seldom been investigated and some species might remain fertile at higher doses.

Understanding the genes involved in tardigrade resistance to IR is essential to unraveling the mechanisms of their exceptional resilience. Systematic comparison of whole genome sequences has suggested that tardigrades have one of the highest proportions of gene gain and gene loss among metazoan phyla (Guijarro-Clarke, Holland, and Paps 2020 [↗](#)). Several novel, tardigrade specific genes have indeed been involved in resistance to desiccation including CAHS, MAHS, SAHS, and AMNP gene families (Hesgrove and Boothby 2020 [↗](#); Arakawa 2022 [↗](#); Yamaguchi et al. 2012 [↗](#); S. Tanaka et al. 2015 [↗](#); Yoshida et al. 2022 [↗](#)). For resistance to IR, the tardigrade specific gene Dsup (for DNA damage suppressor) has been discovered in *Ramazzottius varieornatus*. Dsup encodes an abundant chromatin protein that increases resistance to X-rays when expressed in human cells (Hashimoto et al. 2016 [↗](#)). *In vitro* experiments have shown that DNA damage induced by hydroxyl radicals was reduced when Dsup was added to nucleosomal DNA (Chavez et al. 2019 [↗](#)), indicating DNA protection by Dsup. However, it is not yet possible to inactivate genes with CRISPR-Cas9 in tardigrades (Goldstein 2022 [↗](#)) and direct evidence for the importance of

Dsup in radio-resistance of tardigrades is still lacking. Interestingly, the presence of resistance genes differs across tardigrade genomes (Arakawa 2022 [↗](#)). While AMNP genes are found in both classes of tardigrades, Heterotardigrada and Eutardigrada, CAHS, SAHS, and MAHS genes are only found in Eutardigrada, and Dsup appears restricted to the Hypsibioidea superfamily of Eutardigrada (Arakawa 2022 [↗](#)). However, given the range of species demonstrated to be radio-resistant across the phylum (Hashimoto and Kunieda 2017 [↗](#)), it seems likely that additional tardigrade specific genes are involved in tardigrades' radio-resistance.

In addition to tardigrades, other animals display exceptional resistance to IR including rotifers, nematodes and larvae of *Polypedium vanderplanki* midges, all surviving doses more than 100 times higher than humans. Recent studies have begun to shed light on the mechanisms involved (Ujaoney et al. 2024 [↗](#)). In addition to DNA protection, DNA repair may also help maintain genome integrity upon irradiation. In rotifers, the rate of DNA double-strand breaks is equivalent to that in human cells (Gladyshev and Meselson 2008 [↗](#)), showing that DNA repair, rather than DNA protection, plays a predominant role in their radio-resistance. Furthermore, it was recently found that genes of DNA repair are upregulated in response to IR in rotifers and *P. vanderplanki* larvae (Moris et al. 2023 [↗](#); Ryabova et al. 2017 [↗](#)). In rotifers, upregulation of a DNA ligase gene acquired by horizontal gene transfer may be essential to radio-resistance (Nicolas et al. 2023 [↗](#)). In prokaryotes, radio resistance has been investigated in the bacterium *Deinococcus radiodurans*, showing highly efficient DNA repair in response to the high levels of DNA damage induced by high doses of IR and the contribution of *D. radiodurans* DNA repair genes (Timmins and Moe 2016 [↗](#)). Previous studies have suggested that upregulation of DNA repair genes may also play a role in radio-resistance of tardigrades: irradiation with IR increases expression of Rad51, the canonical recombinase of homologous recombination (HR), in *Milnesium ineptum* (Beltrán-Pardo et al. 2013 [↗](#)), and regulation of DNA repair genes was observed in *R. varieornatus* (Yoshida et al. 2021 [↗](#)).

To improve our understanding of resistance to IR in tardigrades, we sought to characterize DNA damage and repair after irradiation and to identify novel tardigrade specific genes involved in resistance to IR. For this purpose, we first examined the kinetics of DNA damage and repair after IR in the model species *H. exemplaris*. This species was chosen due to its ease of rearing in laboratory conditions and its known genome sequence. Additionally, to identify novel genes involved in resistance to IR, we analyzed gene expression in response to IR in *H. exemplaris* and two additional species, *Acutuncus antarcticus*, from the Hypsibioidea superfamily (Giovannini et al. 2018 [↗](#)), and *Paramacrobiotus fairbanksi* of the Macrobiotidae superfamily (Guidetti et al. 2019 [↗](#)). Together with multiple DNA repair genes, a tardigrade specific gene, which we named Tardigrade DNA damage Response gene 1 (TDR1) was strongly upregulated in response to IR in all three species analyzed. Further analyses in *H. exemplaris*, including differential proteomics and Western blots, showed that TDR1 protein is present and upregulated. *In vitro* experiments demonstrated that recombinant TDR1 interacts with DNA and forms aggregates with DNA at high concentrations. Importantly, when expressed in human cells, TDR1 reduced the number of phospho-H2AX foci induced by Bleomycin, a DNA damaging drug used as a radiomimetic. These findings show the importance of DNA repair in radio-resistance of tardigrades and suggest that TDR1 is a novel tardigrade specific DNA binding protein involved in DNA repair after exposure of tardigrades to IR.

Results

Double- and single-strand breaks are induced and repaired after exposure of *H. exemplaris* to IR

IR causes a variety of damages to DNA such as nucleobase lesions, single- and double-strand breaks (Téoule 1987). In eukaryotes, from yeast to humans, phosphorylation of H2AX is a universal response to double-strand breaks (DSBs) and an early step in the DNA repair process (Fernandez-Capetillo et al. 2004). To investigate DSBs caused by IR, we generated an antibody against phosphorylated H2AX of *H. exemplaris* (Supp Figure 1). *H. exemplaris* tardigrades were exposed to either 100 or 1000 Gy of ^{137}Cs γ -rays, which are known to be well tolerated by this species (Beltrán-Pardo et al. 2015). We analyzed phospho-H2AX in protein extracts of *H. exemplaris* collected at 30 min, 4h, 8h30, 24h and 73 h after irradiation. For both 100 and 1000 Gy doses, phospho-H2AX was detected at 30 min after irradiation, reached its peak levels at 4h and 8h30 and then gradually decreased (Figure 1a). Irradiation was also performed with an accelerated electron beam, which delivered identical doses in much shorter times, 1000 Gy in 10 min instead of 1h for the ^{137}Cs source, in order to better appreciate the early peak of phospho-H2AX. A peak of phospho-H2AX was detected at 4h and a similar, gradual decrease was observed (Supp Figure 2a). Next, we performed whole mount immunolabeling of tardigrades and observed intense, ubiquitous phospho-H2AX labeling in nuclei 4h after 100 Gy irradiation, which had significantly decreased 24h later (Figure 1b). This suggests irradiation impacts all adult cells and indicates efficient DNA repair by 24h after 100 Gy irradiation, consistent with the results of Western blot analysis. After 1000 Gy irradiation, the intense signal detected at 4h had decreased in most nuclei at 24h but it persisted at high intensity specifically in gonads (Supp Figure 2b-c). The finding of persistent DSBs in gonads at 72h after 1000 Gy likely explains why *H. exemplaris* no longer lay eggs and become sterile after exposure to 1000 Gy (Beltran-Pardo 2005). In order to investigate DNA synthesis taking place after irradiation, we incubated tardigrades with the thymidine nucleotide analog EdU (Gross et al. 2018). Using confocal microscopy, we could detect DNA synthesis in replicating intestinal cells of control animals, as previously shown by (Gross et al. 2018). In contrast, we could not detect any specific signal in irradiated tardigrades compared to controls, suggesting (i) that DNA synthesis induced during DNA repair remained at low, undetectable levels and (ii) that dividing intestinal cells detected in control animals were irreversibly damaged by the 1000 Gy irradiation (Supp Figure 2d). Together, these results demonstrate the dose-dependent induction and repair of DSBs in response to IR. Phospho-H2AX immunolabeling experiments also suggested that 1000 Gy induces irreversible damage in the gonads and dividing intestinal cells.

Next, we assessed the physical integrity of genomic DNA at several time points after irradiation. Samples from Figure 1a were run in native agarose gels and irradiated samples were found to be indistinguishable from non-irradiated controls (Supp Figure 2e), showing DSBs and the resulting DNA fragmentation could not be detected in this experimental setting. Single-strand breaks (SSBs) were evaluated by migrating DNA samples in denaturing agarose gels (Figure 1c). DNA from control, untreated tardigrades appeared as a predominant band running above the 20 kb marker with a smear. The smear, likely due to the harsh extraction conditions needed for tardigrade cuticle lysis, extended down between 20 and 10 kb markers where a discrete band, of unknown origin, could be detected (Figure 1c). At 30 min after 1000 Gy irradiation, intensity of the high molecular weight band was drastically reduced, and DNA detected in the smear between 10 and 20 kb was strongly increased. In addition, the discrete band could no longer be detected. This clearly indicates that 1000 Gy IR induces high rates of SSBs. Considering that the majority of DNA fragments detected had a size of 10 to 20 kb and that the discrete band of 10-20 kb could no longer be detected, we can roughly evaluate that there is approximately 1 SSB every 10 to 20 kb. This corresponds to induction of SSBs at a rate of 0.05-0.1 SSB/Mb/Gy. Between 4h and 24h, the

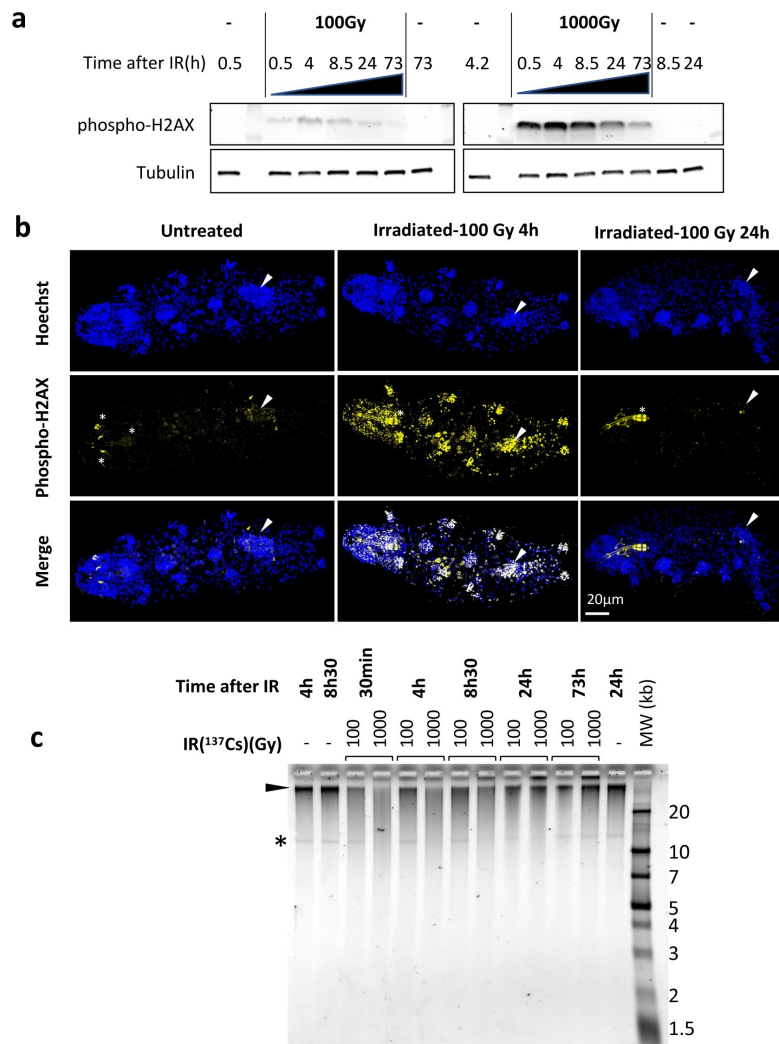


Figure 1

Analysis of DNA damage and repair in *Hypsibius exemplaris* after γ-ray irradiation.

(a) Analysis of phospho-H2AX expression after exposure of *H. exemplaris* to IR. Western blot analysis with in-house antibody against phosphorylated *H. exemplaris* H2AX (anti-phospho-H2AX) at indicated time points after irradiation of tardigrades with indicated dose of γ-ray irradiation. Phospho-H2AX levels were normalized by total α-tubulin expression levels and quantification is provided in Supp Figure 2a. (-) lanes show extracts from control tardigrades processed in parallel to irradiated tardigrades at indicated time points post-irradiation.

(b) Analysis of phospho-H2AX expression in whole mount *H. exemplaris* after exposure to 100 Gy. Tardigrades were exposed to 100 Gy, fixed with 4% PFA at 4 and 24h post irradiation, immunolabeled with anti-phosphoH2AX antibody and anti-rabbit IgG conjugated to Alexa488 and visualized by confocal microscopy using the Airyscan2 module. Maximum projection of confocal Z-stack are shown. Images at different time points were taken with identical settings so that signal intensity could be compared. Upper panel shows Hoechst staining of nuclei (in blue). Arrowhead indicates position of the gonad (revealed by intense Hoechst and larger nuclei signal). The gonad exhibits intense labeling phospho-H2AX at 4h which is no longer detected at 24h, showing efficient DNA repair consistent with preservation of the capacity to lay eggs and reproduce after 100 Gy IR (Beltran-Pardo et al, 2005). * indicates autofluorescence of bucco-pharyngeal apparatus. Scale bar 20μm

(c) Analysis of single-strand breaks by denaturing agarose gel electrophoresis of DNA isolated from ~8000 *H. exemplaris* at indicated time points post-irradiation (100Gy or 1000Gy γ-rays from ¹³⁷Cs source). (-) indicates DNA from control, non-irradiated tardigrades collected and processed in parallel to treated samples from indicated time points. MW corresponds to the Molecular Weight ladder. * indicates a discrete band of single stranded DNA detected in *H. exemplaris* genomic DNA. Arrowhead indicates high molecular weight single-stranded DNA that is not resolved by agarose gel electrophoresis. (-) lanes show DNA from control tardigrades processed in parallel to irradiated tardigrades at 4h or 8h30 post-irradiation as indicated.

DNA migration profile was progressively restored and between 24h and 73h, it was identical to controls. Similar results were observed with the 100 Gy dose (**Figure 1c**). However, compared to 1000 Gy, the changes observed were not as marked and the discrete 10-20 kb band could always be detected, indicating SSBs were induced at lower rates. These results indicate that SSBs are inflicted by IR in a dose-dependent manner, roughly estimated to 0.05-0.1 SSB/Mb/Gy, and progressively repaired within the next 24 to 73h (**Figure 1c**).

***H. exemplaris* strongly overexpresses canonical DNA repair genes as well as RNF146 and TDR1, a novel tardigrade specific gene, in response to IR**

To examine the gene expression changes associated with tardigrade response to IR, we performed RNA sequencing of *H. exemplaris* collected 4h after irradiation. The analysis revealed that 421 genes were overexpressed more than 4-fold (with an adjusted p-value < 0.05) including 120 overexpressed more than 16-fold (**Figure 2a**, Supp Table 1, Supp Figure 3a). The Gene Ontology analysis of overexpressed genes highlighted a strong enrichment of DNA repair genes (**Figure 2a**, Supp Figure 4). In particular, genes for both major pathways of DNA double strand break repair, HR and NHEJ, were among the most strongly stimulated genes. Examples are genes for RAD51 and MACROH2A1 in HR (Khurana et al. 2014; Baumann and West 1998) and XRCC5 and XRCC6 in NHEJ (**Figure 2a**). The gene for POLQ, the key player of the alternative end joining pathway of DNA double strand break repair (Mateos-Gomez et al. 2015), was also strongly upregulated (**Figure 2a**). Also notable among most strongly overexpressed genes were genes for XRCC1, PNKP and LIG1 in Base Excision repair (Whitehouse et al. 2001; Krokan and Bjørås 2013), along with genes for PARP2 and PARP3, which catalyze PARylation of many DNA repair proteins (Pascal 2018) and RNF146 (**Figure 2a**). Interestingly, RNF146 is a ubiquitin ligase that has been reported to be important for tolerance to IR in human cells by targeting PARylated XRCC5, XRCC6 and XRCC1 for degradation (Kang et al. 2011). Our results suggest that RNF146 upregulation could contribute to the remarkable resistance of tardigrades to IR.

Among overexpressed genes, we also observed AMNP gene family members (Yoshida et al. 2022) (one representative was labeled AMNP-like, **Figure 2a**). AMNP genes encode recently discovered tardigrade specific Mn-Peroxidases which are overexpressed in response to desiccation and UVC in *R. varieornatus* (Yoshida et al. 2022). AMNP gene g12777 was shown to increase tolerance to oxidative stress when expressed in human cells (Yoshida et al. 2022). Based on our results, it is possible that AMNP genes such as the AMNP-like gene identified here could contribute to resistance to IR by increasing tolerance to the associated oxidative stress.

In parallel, we also determined the transcriptomic response of *H. exemplaris* to Bleomycin, a well-known radiomimetic drug (Bolzán and Bianchi 2018) (**Figure 2b**, Supp Table 2, Supp Figure 3b). In preliminary experiments, we found that *H. exemplaris* tardigrades survived for several days in the presence of 100 µM Bleomycin, suggesting that *H. exemplaris* could resist chronic genotoxic stress. We hypothesized that key genes of resistance to acute genotoxic stress induced by IR would also be induced by Bleomycin treatment. As expected, the correlation between highly expressed genes after IR and after Bleomycin treatment (with baseMean > 500, Supp Figure 3 and Supp Table 3) was strong for most upregulated DNA repair genes such as XRCC5, XRCC6, PARP2, PARP3, XRCC1, LIG4, LIG1 and RNF146 (**Figure 2c**). Importantly, in addition to DNA repair genes, several genes of unknown function were also strongly overexpressed in both conditions and considered as promising candidates for a potential role in resistance to IR. One such gene, which we named TDR1 (for Tardigrade DNA damage Response 1), was chosen for further investigation. ONT long read sequencing and cDNA cloning of TDR1 allowed us to determine the predicted TDR1 protein sequence which is 146 amino acids long (Supp Figure 5, Supp Table 4). We observed that the current genome assembly predicts a partially truncated TDR1 protein sequence, BV898_14257,

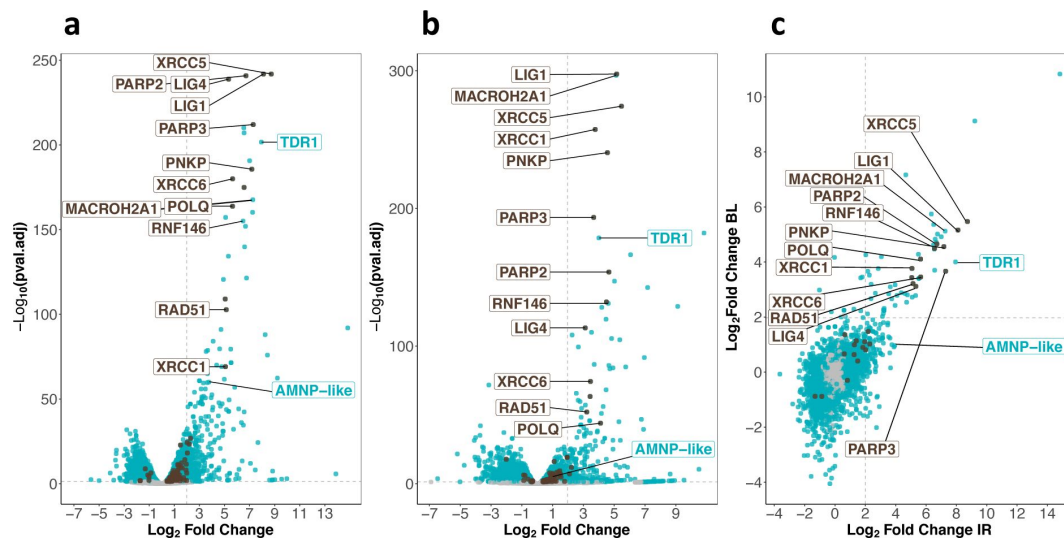


Figure 2

Transcriptomic response of *Hypsibius exemplaris* to IR and Bleomycin.

(a) and (b) Volcano plots representing $\log_2$ Fold Change and adjusted p-value ($-\log$ base 10) of RNA levels between *H. exemplaris* irradiated with 1000Gy γ-rays and untreated controls ($n=3$) (a) and between *H. exemplaris* treated with 100 μM Bleomycin for 4 days and untreated controls ($n=3$) (b). The vertical dotted lines indicate the $\log_2$ Fold Change value of 2 (Fold Change of 4).

(c) Correlation between $\log_2$ Fold Change after exposure to IR and after Bleomycin (BL) treatment for abundant transcripts (with baseMean>500 after DESeq2 analysis). The horizontal and vertical dotted lines indicate the $\log_2$ Fold Change value of 2 (Fold Change of 4).

Blue dots represent transcripts with a $\log_2$ Fold Change with an adjusted p-value $p<0.05$. Brown dots indicate transcripts of DNA repair genes (based on KEGG DNA repair and recombination gene group ko03400) that have a $\log_2$ Fold Change with adjusted p-value $p<0.05$. Grey dots represent transcripts with a $\log_2$ Fold Change with an adjusted p-value $p>0.05$. Brown labels indicate representative strongly upregulated genes of DNA repair. Blue labels indicate two tardigrade specific genes induced in response to IR: the TDR1 gene identified in this work, and the AMNP-like gene (BV898_10264), a member of the family of AMNP/g12777-like genes upregulated in response to desiccation and UVC (Yoshida et al 2022 [\[1\]](#)).

due to an assembly error (Supp Figure 5). Our BLAST analysis against NCBI nucleotide non-redundant database suggested that TDR1 is a novel tardigrade specific gene as no homolog could be found in any other ecdysozoan (Supp Table 5).

Analysis of proteomic response to IR in *H. exemplaris* confirms overexpression of TDR1

We next examined whether stimulation of gene expression at the RNA level led to increased protein levels and in particular, whether TDR1 protein was indeed overexpressed. For this purpose, we first generated specific antibodies to *H. exemplaris* TDR1, XRCC5, XRCC6, and Dsup proteins. Protein extracts from *H. exemplaris* treated with Bleomycin for 4 days or 1000Gy of ψ -rays at 4h and 24h post-irradiation were compared to untreated controls. The apparent molecular weight of the TDR1 protein detected on Western blots was consistent with the expected 16 kD predicted from the 146 amino acid long sequence (**Figure 3a**). Remarkably, similar to phospho-H2AX, TDR1 was only detected after the induction of DNA damage (**Figure 3a**). XRCC5 and XRCC6 protein levels were also stimulated by both Bleomycin and IR treatments, although the fold stimulation was much lower than at the RNA level (**Figure 3a**, Supp Figure 6). Furthermore, we checked expression of *He*-Dsup homolog in *H. exemplaris* (Chavez et al. 2019), which remained constant at the RNA level (see BV898_01301, Supp Tables 1-2), and found that it also remained stable at the protein level after the induction of DNA damage (**Figure 3a**).

To ensure that the observed stimulation was due to new protein synthesis, we treated tardigrades with the translation inhibitor cycloheximide before irradiation (Supp Figure 7a). As expected, no increase in TDR1, XRCC5 or XRCC6 protein levels could be detected after irradiation in extracts from animals treated with cycloheximide (Supp Figure 7b). In particular, TDR1 protein could not be detected when animals were treated with cycloheximide, further confirming that TDR1 is strongly overexpressed in response to IR.

To further extend the analysis of the protein-level response to IR, we conducted an unbiased proteome analysis of *H. exemplaris* at 4h and 24h after irradiation and after Bleomycin treatment using mass spectrometry-based quantitative proteomics. More than 5600 proteins could be detected in all conditions (Supp Table 6). Among them, 58, 266 and 185 proteins were found to be differentially abundant at 4h post-irradiation, 24h post-irradiation and after Bleomycin treatment, respectively compared to control tardigrades ($\text{Log}_2\text{Fold Change} > 0.3$ and limma p-value < 0.01 , leading to a Benjamini-Hochberg FDR $< 3\%$, **Figure 3b**, **Table 1** and Supp Table 6). We observed a good correlation between stimulation at RNA and protein levels (**Figure 3c**). It is worth noting that the fold changes observed for proteins were smaller than those obtained for mRNAs, possibly due to the use of an isobaric multiplexed quantitative proteomic strategy known to compress ratios (Hogrebe et al. 2018). For strongly overexpressed canonical DNA repair genes discussed above, we confirmed significantly increased protein levels in response to IR (**Figure 3b**). RNF146, in contrast, could not be detected, likely due to limited sensitivity of our mass spectrometry-based quantitative proteomics. Importantly, despite the small size of the predicted TDR1 protein, we detected four different TDR1-related peptides, providing direct evidence of strong TDR1 over-expression in response to IR (Supp Table 6).

Conservation of TDR1 and transcriptional response to IR in other tardigrade species

To gain insight into the importance of the upregulation of TDR1 and DNA repair genes in resistance to IR, we chose to investigate its conservation in other tardigrade species. We successfully reared two other species in the lab: *Acutuncus antarcticus*, from the Hypsibioidae superfamily, known for its resistance to high doses of UV, likely related to its exposure to high levels of UV in its natural Antarctic habitat (Giovannini et al. 2018), and *Paramacrobiotus*

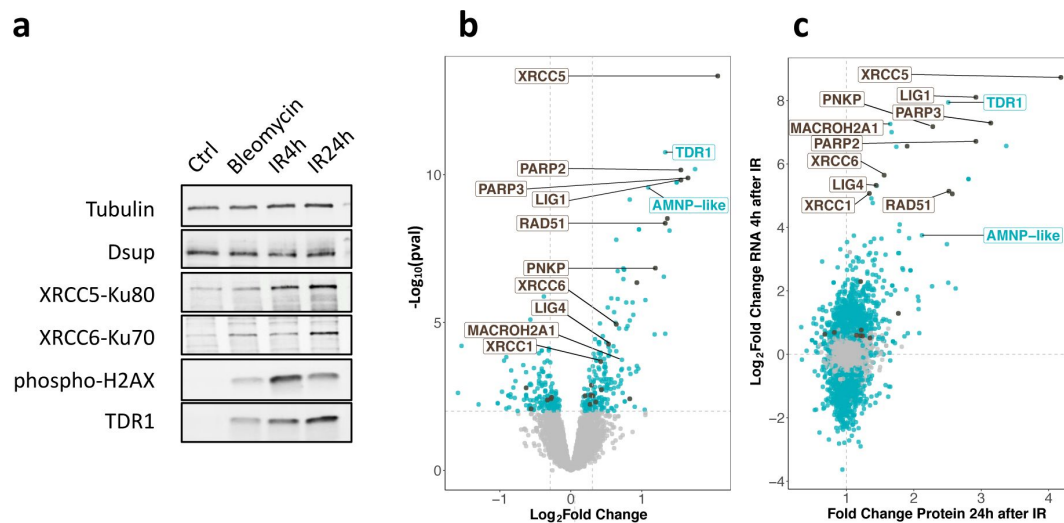


Figure 3

Changes in protein expression in *Hypsibius exemplaris* after exposure to IR.

(a) Western blot analysis of *He*-TDR1, *He*-XRCC5, *He*-XRCC6 (among the most strongly stimulated genes at the RNA level) and *He*-Dsup (not stimulated at the RNA level) in irradiated *H. exemplaris* tardigrades Control, untreated *H. exemplaris* (Ctrl) or *H. exemplaris* treated with 100 μ M Bleomycin for 4 days, or with 1000Gy γ -rays and extracts prepared at indicated times post-irradiation (IR4h and IR24h). Alpha-tubulin was used for normalization and phospho-H2AX for showing induction of DNA double-strand breaks. Quantification of 4 independent experiments are shown in Supp Figure 6c. Molecular weight marker present in uncropped Western blots (Supp Figure 16b) is consistent with the expected 16kDa size of TDR1.

(b) Volcano plot representing Log₂ Fold Change and -Log₁₀(limma p-value) of proteins between *H. exemplaris* 24h post-irradiation with 1000Gy γ -rays and untreated control animals ($n=4$). Blue dots represent proteins with a Log₂ Fold Change with a -Log₁₀(limma p-value) ≥ 2 . Brown dots represent DNA repair proteins (based on KEGG DNA repair and recombination gene group ko03400) with -Log₁₀(limma p-value) ≥ 2 . Grey points represent proteins with Log₂ Fold Change with -Log₁₀(limma p-value) < 2 and the vertical grey lines delimit Log₂(FC) > 0.3 or < -0.3 . Brown labels indicate representative strongly upregulated genes of DNA repair. Blue labels indicate two tardigrade specific genes induced in response to IR: the TDR1 gene identified in this work, and the AMNP-like gene (BV898_10264), a member of the family of AMNP/g12777-like genes upregulated in response to desiccation and UVC (Yoshida et al 2022).

(c) Correlation between Fold Changes of protein levels 24h post-irradiation with 1000 Gy (as measured in (b)) and Log₂FoldChange of RNA levels 4h post-irradiation (as measured in Figure 2a).

1000 Gy γ Irradiation vs Control	<i>IR4h vs Ctrl</i>	<i>IR24h vs Ctrl</i>	<i>Bleomycin vs Ctrl</i>
Total number of identified proteins		5625	
DE Proteins	58	266	185
DE Proteins Up	42	168	128
DE Proteins Down	16	98	57
<i>DE proteins in the 3 conditions</i>		<i>36</i>	
Tardigrade specific DE proteins	13	61	70
Tardigrade specific DE proteins Up	11	52	47
Tardigrade specific DE proteins Down	2	9	23
<i>Tardigrade specific DE proteins in the 3 conditions</i>		<i>9</i>	

Table 1

Proteomic analysis metrics: numbers of differentially expressed (DE) proteins (with limma p-value < 0.01 and Log₂FoldChange<-0.3 or >0.3) for each indicated condition in *Hypsibius exemplaris*.

The numbers of tardigrade specific DE proteins are also indicated. 9 tardigrade specific DE proteins were common to the 3 conditions, the corresponding list is provided in Supp Table 10. Tardigrade specific proteins are defined as detailed in the Methods section.

fairbanksi (Guidetti et al. 2019 [↗](#)), which was reared from a garden moss and was of high interest as a representative of Macrobiotioidea, a major tardigrade superfamily considered to have diverged from Hypsibioidea more than 250 My ago (Regier et al. 2005 [↗](#)). It was in *Paramacrobiotus areolatus*, which also belongs to Macrobiotioidea, that the first demonstration of resistance to IR was carried out (with a LD50_{24h} of 5700Gy) (May 1964 [↗](#)). Importantly, species of Macrobiotioidea examined so far lack Dsup homologs (Arakawa 2022 [↗](#)). We found by visual inspection of animals after IR that *A. antarcticus* and *P. fairbanksi* readily survived exposure to 1000 Gy. As done above in *H. exemplaris*, we therefore examined genes differentially expressed 4hrs after 1000 Gy IR. In both species, we found numerous genes to be significantly overexpressed in response to IR, and similar to what we observed in *H. exemplaris*, upregulation was often remarkably strong (**Figure 4a** [↗](#) and **4b** [↗](#), **Table 2** [↗](#), Supp Tables 7-9, Supp Figure 8). Crucially, we identified TDR1 homologs in transcriptomes of *A. antarcticus* and *P. fairbanksi* and just like in *H. exemplaris*, these TDR1 homologs were among the most overexpressed genes in both species after IR and in response to Bleomycin treatment of *A. antarcticus* (**Table 2** [↗](#), Supp Table 7-9), strongly suggesting a conserved role of TDR1 in resistance to IR. In contrast, as expected from previous studies, we could identify a Dsup homolog in *A. antarcticus* (Aant_geneID_rb_14333, Supp Table 7-8), from the Hypsibioidea superfamily, but not in *P. fairbanksi* from Macrobiotioidea.

Furthermore, similar to *H. exemplaris*, Gene Ontology analysis of overexpressed genes highlighted a robust enrichment of DNA repair genes in *A. antarcticus* and *P. fairbanksi* in response to IR (Supp Figure 9a-b). Notably, a high proportion of genes of the main repair pathways of DNA damages caused by IR (DSB and SSB repair, and Base Excision repair) were significantly overexpressed after IR in all three species (Supp Figure 9c-d) and as in *H. exemplaris*, among the genes with the strongest overexpression in *A. antarcticus* and *P. fairbanksi*, we observed the canonical DNA repair genes for XRCC5, XRCC6, XRCC1, PARP2, PARP3 as well as the gene for RNF146. Interestingly, a set of 50 putative orthologous genes was upregulated in response to IR in all three species, suggesting a conserved signaling and transcriptional program is involved in the response to IR between the distantly related Hypsibioidea and Macrobiotioidea superfamilies (Supp Figure 10).

***He*-TDR1 interacts directly with DNA *in vitro* and co-localizes with DNA in transgenic tardigrades**

In addition to the three species studied, BLAST searches against recent tardigrade transcriptomes enabled the identification of potential TDR1 homologs in other tardigrade species, which all belong to the Macrobiotioidea superfamily (**Figure 5a** [↗](#), Supp Table 4). The TDR1 proteins are predicted to be 146 to 291 amino acids long, with the C-terminal part showing the highest similarity (**Figure 5a** [↗](#)). Interestingly, TDR1 proteins contain a relatively high proportion of basic amino acid residues (20.5% of K or R amino acids for TDR1 of *H. exemplaris*, *He*-TDR1), including at conserved positions in the C-terminal domain (**Figure 5a** [↗](#)). This led us to wonder if TDR1 might interact directly with DNA. To investigate this possibility, we purified recombinant *He*-TDR1 (Supp Figure 11) and tested its interaction with DNA using gel shift assays. As shown in **Figure 5b-c** [↗](#), when circular or linear plasmid DNA was incubated with increasing concentrations of *He*-TDR1, a shift in plasmid mobility was detected in agarose gel electrophoresis, indicating the formation of a complex between *He*-TDR1 and DNA. The observed binding of *He*-TDR1 at a ratio of 1 *He*-TDR1 protein to every 3 bp of DNA is similar to the binding reported for non-sequence specific DNA binding proteins such as the Rad51 recombinase (Zaitseva, Zaitsev, and Kowalczykowski 1999 [↗](#)). Upon adding the highest amounts of *He*-TDR1, we noted that the amount of plasmid DNA detected by ethidium bromide staining appeared to decrease. We ruled out that plasmid DNA was degraded during incubation by performing Proteinase K treatment which revealed that the amounts of intact plasmid DNA had not changed after incubation with *He*-TDR1. As an alternative explanation, we considered that at high *He*-TDR1 concentrations, *He*-TDR1 and DNA might form aggregates that could not enter the gel. To explore this possibility, we examined mixes of *He*-TDR1-GFP and plasmid DNA by fluorescence microscopy. At ratios at which complex formation was

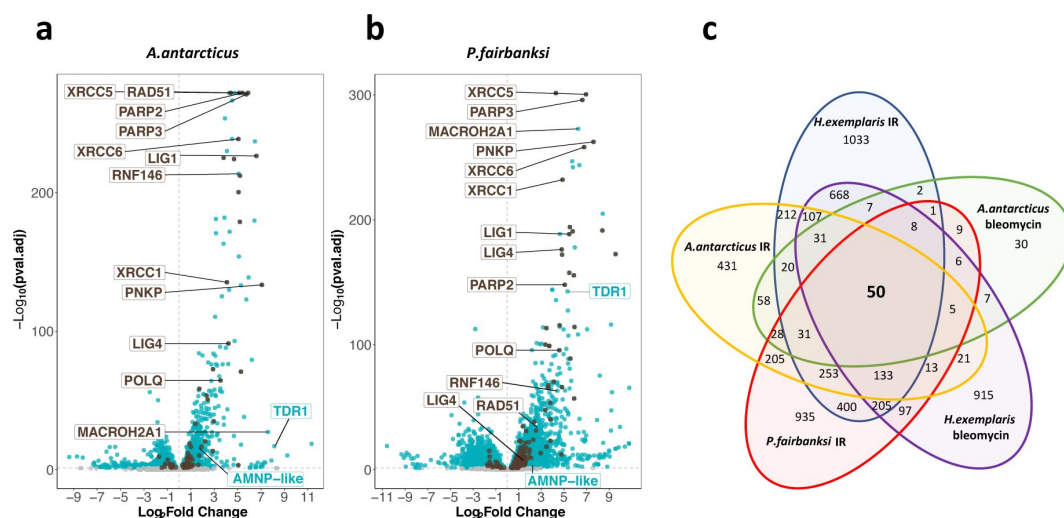


Figure 4

Transcriptomic response of *Acutuncus antarcticus* and *Paramacrobiotus fairbanksi* to IR

(a) and (b) Volcano plots representing Log₂Fold Change and adjusted p-value (–log base 10) of RNA levels after irradiation with 1000 Gy ψ -rays between irradiated *A. antarcticus* and untreated controls ($n=3$) (a) and between irradiated *P. fairbanksi* and untreated controls ($n=3$) (b).

Blue dots represent transcripts with an adjusted p-value p<0.05. Brown dots indicate transcripts of DNA repair genes (based on KEGG DNA repair and recombination gene group ko03400) with an adjusted p-value p<0.05. Brown labels indicate representative strongly upregulated genes of DNA repair. Blue labels indicate two tardigrade specific genes induced in response to IR: the TDR1 gene identified in this work, and the AMNP-like gene (BV898_10264), a member of the family of AMNP/g12777-like genes upregulated in response to desiccation and UVC (Yoshida et al 2020).

(c) Venn diagram showing upregulated genes with an adjusted p-value p<0.05 common to the transcriptomic response to IR in the three species analysed and to Bleomycin in *H. exemplaris* and *A. antarcticus*.

γ Irradiation vs Control	<i>H. exemplaris</i>	<i>A. antarcticus</i>	<i>P. fairbanksi</i>	Bleomycin vs Control	<i>H. exemplaris</i>	<i>A. antarcticus</i>
Total number of DEG	6209	3708	7515	Total number of DEG	5116	1458
DEG Up	3178	1875	3687	DEG Up	2284	399
DEG Down	3031	1833	3828	DEG Down	1113	1059

Table 2

Number of differentially expressed genes (DEG with adjusted p-value < 0.05) after IR with 1000 Gy ψ -rays vs untreated in 3 species (*H. exemplaris*, *A. antarcticus*, *P. fairbanksi*) and Bleomycin treatment for 4 or 5 days in *H. exemplaris* and *A. antarcticus*.

A heatmap of the 50 putative orthologous upregulated genes common to all conditions is given in Supp Figure 10.

detected by agarose gel electrophoresis (**Figure 5b** [↗](#) and **5c** [↗](#)), we observed fluorescent spots in the samples, suggesting the presence of large protein-DNA aggregates (of 2-5 μm) likely unable to enter the agarose gels (Supp Figure 12).

To further examine the potential interaction of *He*-TDR1 with DNA *in vivo*, we generated a tardigrade expression plasmid with *He*-TDR1-mNeonGreen cDNA downstream of *He*-*Actin* promoter sequences and introduced it into tardigrade cells using a recently reported protocol (S. Tanaka, Aoki, and Arakawa 2023 [↗](#)). *He*-TDR1-mNeonGreen was easily detected in muscle cells, likely due to high muscle-specific activity of the *He*-*Actin* promoter, and predominantly localized to nuclei, as observed by confocal microscopy (**Figure 5d-e** [↗](#)). Importantly, *He*-TDR1 co-localized with Hoechst staining, suggesting *He*-TDR1 is able to interact with DNA *in vivo*. In summary, these experiments clearly documented interaction of *He*-TDR1 with DNA but also revealed its unexpected ability to compact DNA into aggregates.

Expression of TDR1 proteins diminishes the number of phospho-H2AX foci in human U2OS cells treated with Bleomycin

Next, we aimed to investigate whether the expression of TDR1 could impact the number of phospho-H2AX foci detected upon treatment of human U2OS cells with the radiomimetic drug Bleomycin. When DSBs occur, H2AX is phosphorylated along extended DNA regions near the break and phospho-H2AX foci can be easily detected by immunolabeling, providing a means to indirectly visualize and quantify DSBs in nuclei (Lowndes and Toh 2005 [↗](#)). We designed plasmids for expression of TDR1 proteins from different tardigrade species fused to GFP and transfected them into human U2OS cells. After 48h, we treated cells with 10 $\mu\text{g}/\text{ml}$ Bleomycin to induce DSBs. This allowed us to quantify phospho-H2AX foci in response to Bleomycin by immunolabeling with anti-human phospho-H2AX antibody. As controls, we transfected plasmids expressing either GFP, *Rv*Dsup-GFP or *He*RNF146-GFP. The quantification of phospho-H2AX was carried out in transfected cells (**Figure 6a** [↗](#) and Supp Figure 13a). As previously demonstrated for *Rv*Dsup (Hashimoto et al. 2016 [↗](#)) and as expected from the characterization of human RNF146 (Kang et al. 2011 [↗](#)), expression of *Rv*Dsup-GFP and *He*RNF146-GFP respectively reduced the number of phospho-H2AX foci. This result strongly suggests that *He*RNF146 is a homolog of human RNF146. Moreover, expression of TDR1-GFP fusion proteins from all species tested also significantly reduced the number of phospho-H2AX foci in human cells treated with Bleomycin, supporting the potential role of TDR1 proteins in tardigrade resistance to IR. **Figure 6b** [↗](#) shows that *He*-TDR1-GFP protein was localized in the nucleus of transfected cells, which is consistent with its ability to directly interact with DNA and its nuclear localization after transgenic expression in *H. exemplaris*.

Discussion

Our study aimed to understand the role of DNA repair in the remarkable radio-resistance of tardigrades. We examined the DNA damage and repair mechanisms in the tardigrade species *H. exemplaris* after exposure to ionizing radiation (IR) and performed comparative transcriptomics in three species of the Tardigrada phylum. Our results indicate that DNA repair plays a major role in the radio-resistance of tardigrades compared to human cells and identified the gene for TDR1, a novel DNA-binding protein highly upregulated in response to IR and likely to play an original function in DNA repair.

DNA repair plays a major role in resistance of tardigrades to IR

Using an antibody raised against phosphorylated *He*-H2AX, we could detect DSBs by Western blot and by immunolabeling (**Figure 1** [↗](#)). Our analysis documented dose-dependent DNA damage and repair taking place after exposure to IR. DNA damage could be detected in virtually all nuclei by immunolabeling. However, at 1000 Gy, phospho-H2AX labeling persisted longer than at 100 Gy in

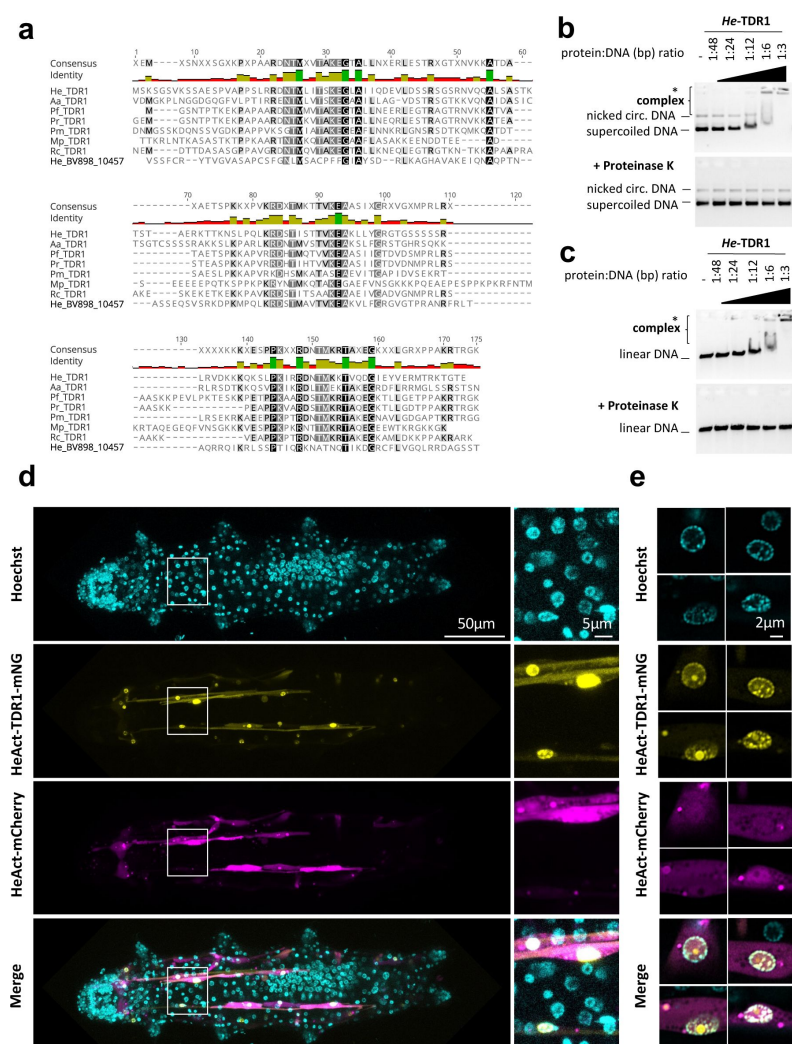


Figure 5

He-TDR1 interacts directly with DNA.

(a) Sequence alignment of the conserved C-terminal domain of TDR1 proteins from *H. exemplaris* (He), *A. antarcticus* (Aa), *P. fairbanksi* (Pf) (identified in this work), and from *P. richtersi* (Pr) (NCBI transcriptome assembly GFGY00000000.1, now known as *P. spatialis*), *P. metropolitanus* (Pm), *R. coronifer* (Rc) (Kamilar et al 2019), *M. philippinus* (Mp) (Mapalo et al 2020). He_BV898_10457 corresponds to a paralog of HeTDR1 in *H. exemplaris* with weaker sequence identity to HeTDR1 than TDR1 homologs from other species.

(b-c) Gel shift assay of recombinant He-TDR1 with circular plasmid (b) or linear plasmid (c). Mixes of plasmid DNA and recombinant He-TDR1 at indicated protein to DNA (bp) ratios were incubated at 25°C for 20 minutes and migrated, either directly or after proteinase K digestion, at room temperature on 0.75% agarose with ethidium bromide. Fluorescence was revealed with a ChemiDoc MP imager. Complexes of plasmid DNA and recombinant He-TDR1 are indicated by a bracket. High-molecular weight complexes that remained in the loading wells and did not migrate into the gel are indicated by an asterisk.

(d) Expression of He-TDR1-mNeonGreen in transient transgenic *H. exemplaris* tardigrades. Expression plasmids of He-TDR1-mNeonGreen (mNG) and mCherry (both under control of the He-Actin promoter) were microinjected into the body fluid of *H. exemplaris* adults and electroporation was performed to induce delivery into cells following the protocol of Tanaka et al. 2023. Confocal microscopy was carried out on live animals immobilized in carbonated water at day 8 post-microinjection after 2 days of treatment with 20 μM Hoechst 33342 to stain nuclei. Maximum projections of confocal Z-stack are shown.

(e) High resolution imaging of nuclei expressing He-TDR1-mNG and Hoechst staining of the nucleus using the Airyscan2 module (one Z-slice is shown). Nuclear He-TDR1-mNG is co-localized with Hoechst staining except for 1 big foci which was observed in some high-resolution images (e-yellow channel), likely corresponding to nucleolar accumulation of overexpressed He-TDR1-mNG.

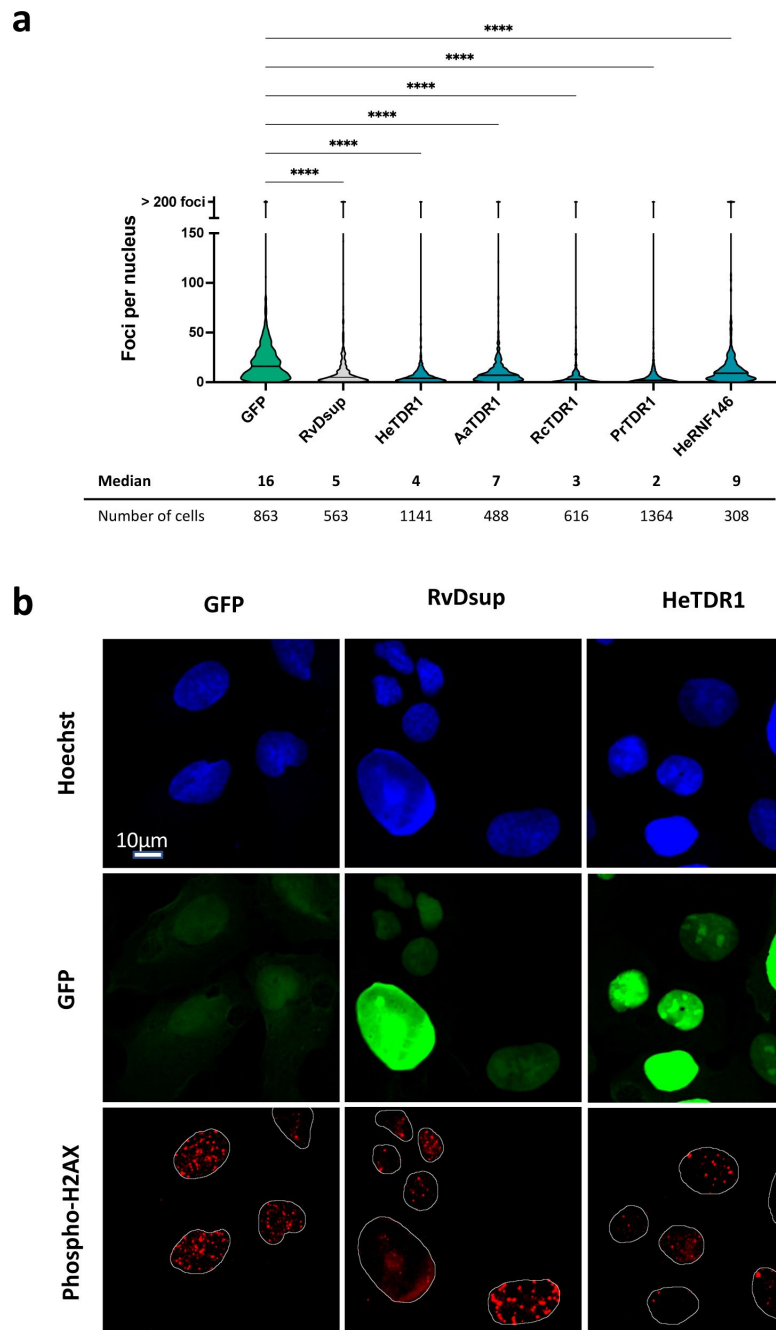


Figure 6

Reduced numbers of phospho-H2AX foci after Bleomycin treatment in human U2OS cells expressing TDR1-GFP from multiple tardigrade species.

(a) Violin plot of the number of phospho-H2AX foci per nucleus of cells expressing the indicated protein. Phospho-H2AX foci were counted after $1\mu\text{g}.\text{ml}^{-1}$ Bleomycin 1 hour-treatment of U2OS cells electroporated with a plasmid expressing either eGFP (control), RvDsup-GFP, TDR1-GFP from *H. exemplaris* (He), *A. Antarcticus* (Aa), *R. coronifer* (Rc) and *P. richtersi* (Pr), He-RNF146-GFP. Cells were fixed with a 4% PFA PBS solution for 1h, immunolabeled with chicken anti-GFP and mouse anti-phospho-H2AX antibodies and imaged by confocal microscopy.

**** indicates $P < 0.0001$ (Kruskal-Wallis test). A minimum of 308 nuclei were counted in each experimental condition ($n=3$). A representative experiment is shown here. Data from independent replicates are given in Supp Figure 13.

(b) Representative confocal fluorescence imaging of experiment analyzed in (a). Images were taken with identical settings and printed with same thresholding so that signal intensity could be compared. Scale bar corresponds to $10\mu\text{M}$.

the gonad. Additionally, at 1000 Gy, cell divisions could no longer be detected in the midgut of the digestive system. These two consequences of exposure to higher doses of IR may be due to higher sensitivity of replicating cells to IR and explain why *H. exemplaris* tardigrades no longer lay eggs and become sterile after irradiation with 1000 Gy (Beltrán-Pardo et al. 2015 [↗](#)).

Using standard agarose gel electrophoresis, we were able to observe that SSBs were induced every 10 to 20 kb in *H. exemplaris* after exposure to 1000 Gy of γ -rays, indicating a rate of 0.05 to 0.1 SSB/Gy/Mb (**Figure 1c** [↗](#)). Remarkably, this rate is roughly similar to that reported for cultured human cells which is 0.17 SSB/Gy/Mb (Mohsin Ali et al. 2004 [↗](#)), showing that high levels of DNA damage are induced after high doses of IR and thus supporting the importance of DNA repair in the radio-resistance of *H. exemplaris* compared to human cells. In radio-resistant rotifers, the rate of DSBs was comparable to non-radioresistant organisms (Gladyshev and Meselson 2008 [↗](#)), also suggesting the importance of DNA repair in radio-resistance. Concerning the role of DNA protection in radio-resistance, further studies would be necessary; in particular, determining the rate of DNA double-strand breaks and testing the importance of Dsup in live *H. exemplaris*. Quantification of phospho-H2AX foci is frequently used as a proxy but given the small size of tardigrade nuclei, standard imaging by confocal microscopy was not sufficient to clearly identify and quantify independent phospho-H2AX foci. Recent developments in super-resolution microscopy could make it possible to perform such quantification in the future (Tillberg and Chen 2019 [↗](#)). Pulse field electrophoresis is a method that would allow to directly examine DNA damage, but it would require to disrupt the cuticle and release DNA without causing damage which would confound the analysis of DNA integrity.

Fine regulation of scaffolding proteins to cope with high rates of DNA damage

We performed comparative transcriptomics in three different species and uncovered the conserved upregulation of a wide number of DNA repair genes in response to IR (**Figure 4** [↗](#) and Supp Figure 9). Remarkably, the strongest upregulations, both at RNA and protein levels, were detected for proteins acting early in DNA repair in the different pathways involved: XRCC5/XRCC6 in NHEJ, POLQ in MMEJ, XRCC1 in SSB, and PARP2/PARP3 which act as DNA damage sensors common to all double-strand break repair pathways (Pandey and Black 2021 [↗](#)). These early acting proteins either stabilize DNA ends or provide essential scaffolding for subsequent steps of DNA repair. It is possible that producing higher amounts of such proteins is essential to maintain DNA ends long enough for more limiting components of DNA repair to cope with an exceptionally high number of damages. For XRCC5 and XRCC6, our study established, by two independent methods, proteomics and Western blot analyses, that the stimulation at the protein level was much more modest (6 and 20-fold at most, Supp Figure 6) than at the RNA level (420 and 90 fold respectively). This finding suggests that the abundance of DNA repair proteins does not simply increase massively to quantitatively match high numbers of DNA damages. Interestingly, in response to IR, the RNF146 ubiquitin ligase was also found to be strongly upregulated. RNF146 was previously shown to interact with PARylated XRCC5 and XRCC6 proteins and to target them for degradation by the ubiquitin-proteasome system (Kang et al. 2011 [↗](#)). To explain the more modest fold stimulation of XRCC5 and XRCC6 at protein levels compared to its massive increase at mRNA levels, it is therefore tempting to speculate that, XRCC5 and XRCC6 protein levels (and perhaps that of other scaffolding complexes of DNA repair) are regulated by a dynamic balance of synthesis, promoted by increased gene transcription, and degradation, made possible by RNF146 upregulation. Consistent with this hypothesis, we found that, similar to human RNF146 (Kang et al. 2011 [↗](#)), *He*-RNF146 expression in human cells reduced the number of phospho-H2AX foci detected in response to Bleomycin (**Figure 6** [↗](#)).

Further studies should investigate the molecular mechanisms leading to such marked upregulation of RNA levels of these genes. Functional analysis of promoter sequences in transgenic tardigrades is now possible (S. Tanaka, Aoki, and Arakawa 2023 [↗](#)) and could help to

identify a conserved set of transcription factors and/or co-regulators common to Macrobiotodea and Hypsibioidea tardigrades. Such information would provide original insight into the acquisition of resistance to IR and help analyze its relation to the resistance to desiccation. Another outstanding issue, given the high rates of DNA damage taking place, is whether DNA repair is accurate. This is particularly relevant for germ line cells where mutations will be transmitted to the progeny and could impact evolution of the species.

A novel tardigrade specific DNA binding protein involved in resistance to IR

Among the genes overexpressed in response to IR in the three species studied, we identified TDR1 as a promising tardigrade specific candidate (Figures 3 and 4). At the functional level, we found that TDR1 protein interacts with DNA (Figure 5) and that when expressed in human cells, TDR1 protein can reduce the number of phospho-H2AX foci induced by Bleomycin (Figure 6). TDR1 is strongly overexpressed in response to IR (Figure 3), suggesting that it favors DNA repair. Proteins directly involved in DNA repair, however, usually accumulate at sites of DNA damage (Rothkamm et al. 2015), which did not appear to be the case for TDR1 overexpressed in human cells. Given that TDR1 can form aggregates with DNA *in vitro*, we speculate that it may favor DNA repair by regulating chromosomal organization. Intriguingly, the DNA binding activities of TDR1 are reminiscent of DdrC from *D. radiodurans*. DdrC is a small DNA binding protein which is among the most strongly overexpressed proteins after irradiation of *D. radiodurans* with ψ -rays (M. Tanaka et al. 2004) and DdrC forms aggregates with DNA *in vitro* (Banneville et al. 2022; Bouthier de la Tour et al. 2017). Further investigations of TDR1 may thus reveal unexpected parallels between mechanisms of DNA repair conferring radio-resistance in tardigrades and bacteria.

Recent progress in tardigrade transgenesis (S. Tanaka, Aoki, and Arakawa 2023) and promising findings of somatic mutagenesis by CRISPR-Cas9 (Kumagai, Kondo, and Kunieda 2022) are paving the way towards germ line gene editing in *H. exemplaris*. Knocking out Dsup and TDR1 genes should help to better appreciate their importance in radio-resistance and the underlying mechanisms. The C-terminal portion of TDR1 is conserved in species of Macrobiotodea and Hypsibioidea superfamilies of the Parachela order of Eutardigrada but absent from *Milnesium ineptum*, the only representative of the Apochela order of Eutardigrada for which transcriptomic data is currently available (Supp Figure 14), and from Heterotardigrada. Compared to Dsup, which has only been found in Hypsibioidea, TDR1 appears more widely present and could be a more ancient tardigrade gene. As additional tardigrade species, more fully representing the phylogenetic diversity of the phylum, are reared in laboratory conditions and become amenable to experimental analysis, more novel genes and mechanisms of radio-resistance may become apparent. Generally, evolution of tardigrade specific gene sequences appears highly dynamic in the phylum (Arakawa 2022), and further sequencing of tardigrade genomes will help get a better picture of gain and loss of tardigrade specific genes and their relation to resistance to extreme conditions.

In conclusion, our findings suggest that DNA repair is a major contributor to tardigrade radio-resistance. Functional investigations of TDR1, as well as the study of transcriptional regulation in response to IR, will contribute to a deeper understanding of the mechanisms underlying radio-resistance. Additionally, we believe that, as done here, further exploration of tardigrade-specific genes and comparative studies among tardigrade species will shed light on the evolution and diversity of radio-resistance mechanisms in these fascinating organisms.

Material and methods

Tardigrade culture

Hypsibius exemplaris (strain Z151, Sciento, UK) and *Acutuncus antarcticus* (Giovannini et al. 2018 [DOI](#)) were cultivated from 1 individual in mineral water (Volvic, France) at 16°C with a 14h day and 10h night cycle and fed with *Chlorella vulgaris* microalgae (from Algothèque MNHN, France or ordered from *ldc-algae.com* [DOI](#) (Greenbloom pure fresh Chlorella). Microalgae were grown in 1X Cyanobacteria BG-11 Freshwater Solution (C3061, Sigma-Aldrich) at 23°C until saturation, collected by centrifugation at 2000g for 5 min and resuspended in Volvic mineral water 10-fold concentrated before feeding. For concentrated large amounts of *H. exemplaris* 0.2% of linseed oil emulsion (containing 95% linseed oil (organic), 1% Tween 80 (P4780, Sigma-Aldrich) and 4% (+/-)- α -Tocopherol (T3251, Sigma-Aldrich) could be spread at the bottom of the untreated petri dishes before adding algae and water. *Paramacrobiotus fairbanksi* was isolated from a suburban garden moss by A.D.C. and cultivated in the same conditions adding rotifers isolated from the same moss sample (grown separately in Volvic mineral water and fed with *C. vulgaris*) of as supplementary food for adults. Identification as *P. fairbanksi* was achieved by morphological and DNA markers (Supp Figure 15).

IR and Bleomycin treatments of tardigrades

Prior to treatments, tardigrades were separated from *Chlorella* by filtering with a cell strainer 70 μ m mesh (141379C, Clearline) for *H. exemplaris* or 100 μ m mesh (141380C, Clearline) for *A. antarcticus* and *P. fairbanksi*. The cell strainer containing the washed tardigrades was put in a Petri dish containing Volvic mineral water for 3 to 7 days in order to allow live tardigrades to go out of the cell strainer and obtain tardigrades without any remaining *Chlorella*. Tardigrades were then collected on 40 μ m mesh (141378C, Clearline) and washed with Volvic mineral water before proceeding to treatments. For each treatment, tardigrades were collected and split into treated and control samples. Control samples were subjected to the same conditions (temperature, travel (for irradiation), solvent (for Bleomycin)) as the treated tardigrades. For ionizing irradiations, tardigrades were exposed to a ^{137}Cs γ -ray source (GSR-D1 irradiator, RPS Services Limited) at a dose rate of 12.74 or 16 Gy/min for a total dose of 100 or 1000 Gy. Samples were collected at different time points after the end of irradiation (for differential transcriptome analyses, at 4h post-irradiation; for differential proteomics, at 24h post-irradiation; for DNA damage and Western blot analyses, at different time points from 0h to 7 days as indicated in the text). For investigation of early first time point just after ionizing radiation, an electronic beam (KINETRON, GCR-MeV) with 4.5MeV energy and maximum dose rate of 4.103 Gy/sec was also used (Lansonneur et al. 2019 [DOI](#)). For Bleomycin treatment, after separating tardigrades from *Chlorella* by filtration, Bleomycin sulfate (#B5507, SIGMA) was added to the water at a concentration of 100 μ M for 4 or 5 days. The response to Bleomycin treatment was examined in *H. exemplaris* and *A. antarcticus* but not in *P. fairbanksi* tardigrades (which grow more slowly than *H. exemplaris* and *A. antarcticus*).

Production of antibodies against *H. exemplaris* proteins

Antibodies were raised against *H. exemplaris* proteins in rabbits by injecting selected peptide sequences by Covalab (Bron, France). For He-Ku80 (XRCC5), He-Ku70 (XRCC6 C-term), He-Dsup, He-TDR1 two peptides were injected in two rabbits and serums tested by Western blot on *H. exemplaris* extracts from animals treated with 100 μ M Bleomycin for 4 days. Serum showing the best response on day 88 after injections was purified on Sepharose beads coupled to immunogenic peptides. Peptides used were the following, He-TDR1: Peptide 1 (aa 37-52, C-IQDEVLDSSRSGSRNVcoNH₂), Peptide 2 (aa 109-123, C-DKKKQKSLPKIRRDN-coNH₂); He-Ku80 (XRCC5): Peptide 1 (aa 120-135, C-IQFDEESSKKRFAKR-coNH₂), Peptide 2 (aa 444-457, C-LDGKAKDITYQPNDE-coNH₂); He-Ku70 (XRCC6 Cterm): Peptide 1 (aa 182-197, C-

IRPAQFLYPNEGDIRG-coNH₂), Peptide 2 (aa 365-37, C-YDPEGAHTKKRVYEK-coNH₂); *He*-Dsup: Peptide 1 (aa 63-77, C-KTAEVKEKSKSPAKE-coNH₂), Peptide 2 (aa 166-181, C-KEDASATGTNGDDKKE-coNH₂). Production of antibody to *He*-phospho-H2AX is detailed in Supp Figure 1.

Western blot analysis

For each experiment, more than 10 000 *H. exemplaris* tardigrades were irradiated or untreated, and 1000-2000 tardigrades were collected at different time points after irradiation (30min, 4h, 8h30, 24h and 73h), centrifuged at 8000 rpm in 1.5 mL tubes for 5 min and the pellet frozen at -80°C until analysis. Lysis was carried out by sonication for 15 min (15s ON/15s OFF, medium intensity, Bioruptor, Diagenode) at 4°C in 100 µL/5000 tardigrades pellet of the following solution 12mM sodium deoxycholate, 12mM N-Lauryl sarcosine sodium salt, 80mM Tris-HCl pH8.5, 400mM NaCl, 1X cOmplete protease inhibitor (4693116001, Roche) and 1X PhosSTOP (PHOSS-RO, Roche). 0.4 vol of protein gel loading buffer (LDS 4X, Bio-Rad) and 0.1 vol of DTT 1M was added and the mixture heated at 95°C for 5 min before loading onto Any kD Mini-PROTEAN TGX Stain-Free protein gel (4568124, Bio-Rad) and migration in 1x Tris-Glycine-SDS buffer at 200V. Semi-dry transfer of proteins was performed with Transblot Turbo (Bio-Rad) onto nitrocellulose membrane and the membrane was cut half to separately detect proteins > 50 kDa and <50 kDa. Protein detection was done with rabbit primary antibodies diluted 1:1000 or 1:2000 (20-200ng/mL depending on antibody) in TBS-0.1% Tween 20, 5 % BSA, supplemented with 1:10000 dilution of anti-mouse alpha-Tubulin (Clone B-5-1-2; SIGMA) for 1 to 3 h at room temperature. Membranes were washed 3 times for 5 min with 1X TBS 0.1% Tween 20. Secondary antibodies diluted 1:5000 (anti-rabbit Starbright 700 (12004161, Bio-Rad) for specific tardigrade proteins and anti-mouse Starbright 520 (12005867, Bio-Rad) for alpha-tubulin detection) in TBS-1% milk were incubated on membrane for 1h at room temperature. Membranes were washed 3 times for 5 min with 1X TBS 0.1% Tween 20 and unsaturated fluorescent signal was acquired using Chemidoc MP imager (Bio-Rad) with identical settings for samples to be compared within an experiment.

Genomic DNA extraction and analysis

For each experiment, more than 60 000 *H. exemplaris* tardigrades were irradiated or untreated, and 8000-12000 tardigrades were collected at different times after irradiation (30min, 4h, 8h30, 24h and 73h), centrifuged at 8000 rpm in 1.5 mL tubes for 5 min and the pellet frozen at -80°C until analysis. Genomic DNAs (gDNAs) were extracted using the Monarch HMW DNA extraction kit for Tissue from New England Biolabs (NEB) with the following modifications: Lysis buffer was supplemented with proteinase K before proceeding to lysis. Pellets were resuspended in 35 µL of lysis buffer and grinded on ice for 1 minute. This step was repeating twice leading to a final volume of ≈ 125 µL. After grinding, lysis proceeded in three steps: 1) incubation of 15 min at 56°C under gentle agitation (300 rpm), 2) incubation of 30 min at 56°C and 3) incubation of 10 min 56°C after addition of RNase A. Proteinase K and RNase A were added at the concentration recommended by NEB. Proteins were next separated from the gDNA by adding 40 µL of protein separation solution. Samples were next centrifuged (20 minutes, 16000 g, 20°C). gDNA was precipitated with 2 beads and next eluted from the beads with 100 µL of elution buffer. Extracted gDNAs were analysed by electrophoresis on native (0.9% agarose/1X TAE) or denaturing (0.9% agarose/ 30 mM NaOH/ 1 mM EDTA) gels. Electrophoresis conditions were: 2h30min/ 60 V/ 20°C for native gels, and 15 h/18 V/ 20°C for denaturing gel. Native gels were stained with ethidium bromide and denaturing gels with SyBR Green I.

Immunohistochemistry of tardigrades

Immunohistochemistry protocol was derived from (Vladimir Gross and Mayer 2019 [↗](#); V. Gross, Bährle, and Mayer 2018 [↗](#)). 10 000 tardigrades irradiated or untreated were sampled (by batches of 1000 tardigrades) at different time points after irradiation (5min, 4h, 24 or 72h), heated in Volvic mineral water 5 min at 70°C to extend the tardigrade body and directly fixed with 4% formalin (15686, EMS) in 1X PBS-1% Triton-X100 (PBS-Tx) by adding 5X solution. Fixation was carried out

for 1 to 3 h at room temperature. After 1h of fixation tardigrades were pelleted by centrifugation 5min at 8000rpm and kept in 200µL of fixative solution. The cuticle was punctured by sonication using Bioruptor (Diagenode) in 6x1.5mL tubes at a time (5 pulses of 5s ON/ 5s OFF in medium position). After fixation samples were pelleted by centrifugation and washed with 1mL of 1X PBS-1% Triton-X100 3 times (>3h wash) and tardigrades were transferred to a round well 96 plate for transferring tardigrades under the stereomicroscope. Blocking was done in 200µL of 5% BSA in 1X PBS-1% Triton-X100 for at least 1h and in-house primary antibody against phospho-H2AX (rabbit) (dilution 1:10) in blocking buffer was applied on tardigrades overnight or for 3 days at 4°C. Washes with PBS-Tx were done 4 times for several hours the next day transferring tardigrades to a new well filled with 200µL of PBS-Tx. Secondary antibody anti-rabbit Alexa Fluor 488 F(ab')₂ (A11070, Invitrogen, dilution 1:500) in PBS-Tx supplemented with 1% BSA was incubated overnight at room temperature. Washes with PBS-Tx were done 4 times for several hours the next day transferring tardigrades to a new well filled with 200µL. The last wash was done without Triton X100. Hoechst 33342 4µM in PBS 1X was incubated for 30min and tardigrades were quickly transferred to water and to slide with minimum amount of liquid and finally mounted with ProLong™ Glass Antifade Mountant (P36982, Invitrogen).

For analysis of EdU staining, EdU in 20mM Hepes-NaOH pH 7.5 was added to Volvic mineral water with 1000 filtered tardigrades at 50µM 2h before irradiation or with control, untreated tardigrades and kept until 7 days post irradiation. Samples were then processed as in (V. Gross, Bährle, and Mayer 2018 [\[4\]](#)) except for the permeabilization of tardigrades which was carried out by sonication as for phospho-H2AX labelling.

Imaging was done by confocal microscopy (Zeiss (LSM 880 and AiryScan module) with x63 lens) using Zenblack software or Leica DMIRE2 inverted microscope with 10X lens and Metamorph software. Confocal Z-stacks and Maxprojections were processed and adjusted in Fiji ImageJ (v2.9.0). Images were treated with Image J software. Image panels were assembled and labelled in Microsoft powerpoint for Mac (v16.66.1)

RNA sequencing

15000-20000 *H. exemplaris* (n=3), 1000-1500 *A. antarcticus* (n=3) and 500-1000 *P. fairbanksi* (n=4) for each independent biological sample were collected and subjected to IR treatments: (i) control animals non irradiated and extracted 4h post-irradiation and (ii) irradiated animals (with ¹³⁷Cs γ-ray source (GSR-D1 irradiator, RPS Services Limited) at a dose rate of 16Gy/min) extracted 4h post IR. 15000 -20000 *H. exemplaris* and 1000-1500 *A. antarcticus* (3 independent biological samples for each) were also subjected to Bleomycin treatment: (iii) control animals kept for 5 days in Volvic water and (iv) treated with 100µM Bleomycin in Volvic mineral water for 5 days. After treatments, tardigrades were collected and washed by filtration on 40µm nylon mesh and transferred to 1.5mL tubes to pellet by centrifugation at 5 min at 8000 rpm. RNA was extracted using Trizol (15596-026, Invitrogen) and by three freeze-thaw cycles in an ethanol-dry ice bath and mechanical disruption with glass beads and a plastic micro-tube homogenizer at each cycle. Yield was approximately 1µg RNA/1000 tardigrades. Integrity of RNAs was checked on an Agilent 2100 Bioanalyzer with the Eukaryote Total RNA Nano kit and only samples with RNA Integrity Number >6 were sequenced. For *H. exemplaris* RNA samples, single-end (1x75) sequencing (TruSeq Stranded) was done on Illumina NextSeq 500 System. For *A. antarcticus* and *P. fairbanksi* (whose genomes are not available), paired-end (2x150) sequencing (TruSeq Stranded) was performed. In addition to short read RNA sequencing in the different experimental conditions, long read sequencing of a mixture of RNA samples of *A. antarcticus* and of RNA samples of *P. fairbanksi* species were performed with Oxford Nanopore Technology (ONT) to help improve transcriptome assembly. 1D libraries were prepared according to ONT protocol with 1D PCR Barcoding kit and full length non-directional sequencing was performed on PromethION instrument (using Clontech- SMART- Seq v4 Ultra Low Input kit). Basecalling was conducted using guppy version (v6.4.2; parameters: --min_qscore 7 --flowcell FLO-MIN106 --kit SQK-PBK004 --use_quantile_scaling --trim_adapters --detect_primer --trim_primers).

De novo transcriptome assembly

De novo transcriptome assembly was performed using full length cDNA sequences for *A. antarcticus* and *P. fairbanksi*. We used RNA-Bloom (v2.0.0; Ka Ming Nipal, 2023), to assemble the long reads, also using a subset of the produced short reads to correct the contigs. Then we used MMSeqs2 easy-cluster (v; parameters: `--min-seq-id 0.85 -c 0.25 --cov-mode 1`) to cluster together transcript isoforms to a gene (Mirdita, Steinegger, and Söding 2019). We set the minimum sequence identity to 0.85 and minimum coverage to 0.25 for both transcriptomes. Because *P. fairbanksi* is triploid with a high level of heterozygosity, we manually clustered differentially expressed genes that were annotated for the same function by EggNOG (see below). We aligned the isoforms from two or more clusters with the same EggNOG annotation using mafft (version 1.5.0; Katoh and Standley 2013) and we visually inspected the alignments on Geneious Prime (v2023.1). When isoforms from two or more clusters were properly aligning together, they were merged. For *A. antarcticus* and *P. fairbanksi*, we conducted the gene expression analysis using the softwares embedded in the Trinity suite (v2.15.0; Haas et al. 2013). We first mapped RNA-seq reads on the transcriptomes using Salmon (statistics of the mapping of RNAseq reads are given in Supp Table 11; Patro et al. 2017), then we measured differential gene expression using DESeq2 (Love, Huber, and Anders 2014). For *H. exemplaris*, as the genome was available, the gene expression analysis was conducted using Eoulsan workflow version 2.2 (v2.2-0-g4d7359e, build0 build on 764feac4fbd6, 2018-04-17 15:03:09 UTC) (Lehmann et al. 2021). We first mapped RNA-seq reads on the *de novo* transcriptomes using STAR (Dobin et al. 2013), then we measured differential gene expression using DESeq2. The results were plotted using R (v4.2.2) with the ggplot2, ggrepel, and VennDiagram packages. Heatmap was plotted using GraphPad Prism (v9.3.1).

To annotate expressed genes from the three species, we ran EggNOG mapper (v2.1.9) on the assemblies using the “genome” mode (Cantalapiedra et al. 2021). We also annotated all expressed genes through a sequence homology search against *Drosophila melanogaster* (GCF_000001215.4), *Caenorhabditis elegans* (GCF_000002985.6) *Homo sapiens* (GCF_000001405.40), *Hypsibius exemplaris* (GCA_002082055.1), *Paramacrobiotus metropolitanus* (GCF_019649055.1) and *Ramazzottius varieornatus* (GCA_001949185.1). Since *H. exemplaris* genome is annotated, we ran the homology search against the target proteomes using blastp (v. 2.14.0). For *A. antarcticus* and *P. fairbanksi*, we conducted the homology search using the transcript as query (blastx) and as target (tblastn). Only blast hits with an e-value <0.05 were kept as potential homologs.

To identify tardigrade-specific genes, we ran a homology search using Diamond (v2.1.6.160) (Buchfink, Reuter, and Drost 2021) on the complete nr database (Downloaded Apr 12 11:17:28 2023) for each transcript from *A. antarcticus* and *P. fairbanksi* (by blastx) or for each protein sequence for *H. exemplaris* (by blastp). Sequences with no-hit on the nr database (diamond blastx or blastp `-e 0.001 --taxon-exclude 42241 --ultra-sensitive`) and no hit in the previous annotation using proteomes of *C. elegans*, *D. melanogaster* and *H. sapiens* (reciprocal hit – blastx and tblastn) were considered as tardigrade specific and noted “TardiSpe” in Supp Tables and data (Mapalo et al. 2020; Hara et al. 2021; Kamilari et al. 2019). Additionally, for TDR1 we conducted a blast search on the nr database using NCBI blastp, which is more sensitive but slower, than diamond blastp. In contrary to diamond, NCBI blastp produced multiple hits but on non-ecdysozoans organisms only (see Supp Table 5). Similar results were obtained using HMMER (hmmsearch on EMBL-EBI website) on the reference proteome database (see Supp Table 5).

Proteome analysis

For each replicate (n=4 independent biological samples), 18000 tardigrades for each of the four experimental conditions: i) untreated; ii) treated with Bleomycin at 100μM for 4 days; iii) Irradiated (with ¹³⁷Cs γ-ray source (GSR-D1 irradiator, RPS Services Limited) at a dose rate of 12.74Gy/min) and collected after 4h; iv) irradiated and collected after 24h. The tardigrades were split in two samples, with 13 000 tardigrades for differential proteomic analysis and 5000

tardigrades for western blotting experiments, that were pelleted by centrifugation in 1.5mL tubes (8000 rpm for 5 min). The pellets were frozen at -80°C until all samples were available. All samples were lysed the same day 2 weeks before proteomics analysis in 100µL iST-NHS-Lysis buffer (PreOmics GmbH) by sonication (Bioruptor Diagenode, 15s ON/15s OFF for 15 min), and heating at 95°C for 10min. Soluble fractions were collected by centrifugation at 13000g for 15min at 4°C and frozen at -80°C until analysis. Protein concentration in each sample was measured using BCA assay (Sigma-Aldrich). 30µg of each sample were then prepared using the iST-NHS kit (Preomics). Peptides resulting from LysC/trypsin digestion were labelled using TMTpro™ 16plex Label Reagent Set (ThermoFisher Scientific) before mixing equivalent amounts for further processing. The peptide mix was then fractionated using the Pierce High pH Reversed-Phase Peptide Fractionation Kit (ThermoFisher Scientific). The 8 obtained fractions were analyzed by online nanoliquid chromatography coupled to MS/MS (Ultimate 3000 RSLCnano and Q-Exactive HF, Thermo Fisher Scientific) using a 180 min gradient. For this purpose, the peptides were sampled on a precolumn (300 µm x 5 mm PepMap C18, Thermo Scientific) and separated in a 200 cm µPAC column (PharmaFluidics). The MS and MS/MS data were acquired by Xcalibur (version 2.9, Thermo Fisher Scientific). The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE (Perez-Riverol 2022 [PRIDE](#)) partner repository with the dataset identifier PXD043897.

Peptides and proteins were identified and quantified using MaxQuant (version 1.6.17.0, [Cox and Mann 2008](#) [PRIDE](#)) and the NCBI database (*Hypsibius dujardini* taxonomy, 2021-07-20 download, 20957 sequences), the UniProt database (*Chlorella* taxonomy, 2021-12-10 download, 21 219 sequences) and the frequently observed contaminant database embedded in MaxQuant (246 sequences). Trypsin was chosen as the enzyme and 2 missed cleavages were allowed. Peptide modifications allowed during the search were: C6H11NO (C, fixed), acetyl (Protein N-ter, variable) and oxidation (M, variable). The minimum peptide length and minimum number of unique peptides were respectively set to seven amino acids and one. Maximum false discovery rates - calculated by employing a reverse database strategy - were set to 0.01 at peptide and protein levels. Statistical analysis of MS-based quantitative proteomic data was performed using the ProStaR software (Wieczorek et al. 2017 [PRIDE](#)). Proteins identified in the reverse and contaminant databases, proteins identified only in the *Chlorella* database, proteins only identified by site, and proteins quantified in less than three replicates of one condition were discarded. After log2 transformation, extracted corrected reporter abundance values were normalized by Variance Stabilizing Normalization (vsd) method. Statistical testing for comparison of two conditions was conducted with limma, whereby differentially expressed proteins were sorted out using a Log₂(Fold Change) cut-off of 0.3 and a limma p-value cut-off of 0.01, leading to a FDR inferior to 3 % according to the Benjamini-Hochberg estimator.

Production of recombinant *He*-TDR1 and *He*-TDR1-GFP

He-TDR1 and *He*-TDR1-GFP (see plasmid sequence in Supp Table 4) were transformed in *E. coli* Rosetta™ 2(DE3). Singles competent cells (Novagen, MerckMillipore). Protein expression was induced with 1 mM IPTG at OD₆₀₀=0.6-0.7 in 2xYT medium (containing 50 µg/mL carbenicillin, 35 µg/mL chloramphenicol and 1% glucose) at 25°C during 20h. Cells were resuspended in lysis buffer 25 mM Tris-HCl pH 8, 500 mM NaCl, 20 mM imidazole, 1 mM TCEP (supplemented with protease inhibitor cocktail (Roche)), and lysed by sonication (Vibracell 75186 -7 sec ON / 7 sec OFF, 50% amplitude, 10 min). The first step of purification was binding on Ni Sepharose 6 Fast Flow resin in batch (overnight at 4°C). After binding, the resin was washed with lysis buffer and the protein was eluted with 25 mM Tris-HCl pH 8, 500 mM NaCl, 250 mM imidazole, 10% glycerol, 1 mM TCEP. Eluted protein is concentrated (Amicon Ultra 10K) and diluted in buffer 25 mM Tris-HCl pH8, 150 mM NaCl, 10% glycerol, 1 mM TCEP. The second step of purification was a gel filtration Superdex 200 increase 10/300 GL (Cytiva) equilibrated with 25 mM Tris-HCl pH 8, 150 mM NaCl, 10% glycerol, 1 mM TCEP using AKTA Pure instrument (Cytiva). Molecular weight calibration was obtained using Gel Filtration Standard (Bio-Rad).

Protein-DNA interaction assays

For *He*-TDR1 interaction with plasmid DNA, a 5900 bp plasmid (a kind gift of Xie, Kwok, and Scully 2009 [\[4\]](#)) circular or linearized at 20 ng/μL (ie: 30 μM in bp was incubated with increasing amounts (0.625 to 10 μM with 2-fold serial dilutions) of recombinant *He*TDR1 or in buffer containing 15 mM Tris-OAc pH8, NaCl 180 mM, Glycerol 2%, DTT 5 mM, BSA 0.1 mg/mL.

After 20 min binding at room temperature, samples were diluted 2-fold with sucrose 50% (or sucrose 50% with proteinase K 80 U/μL and loaded onto a 0.75% agarose gel containing ethidium bromide. Migration was carried out for 35 min at 100 V (room temperature) and gel was imaged using GBox camera (Syngene).

For imaging of protein-DNA complexes, 1 μL of 5900bp plasmid at 200 ng/μL was added to 10 μL of 10 μM of *He*TDR1-GFP in 10 mM Tris-HCl PH8, 150 mM NaCl, 10% Glycerol and 1 mM TCEP (protein storage buffer) to allow 30 μM in bp (i.e.: 5 nM in plasmid molecule) final concentration. After 10 min incubation at room temperature the reaction was observed in a Kova counting chamber using Leica DMIRE2 40X lens. Images were acquired using Coolsnap HQ camera run by Metamorph software and treated with ImageJ software.

Expression of *He*-TDR1-mNeongreen in *H. exemplaris* tardigrades

Act-*He*-TDR1-mNeongreen (NG) and Act-mCherry expression plasmids were constructed by Gibson assembly with plasmid backbone from (Loulier et al. 2014 [\[4\]](#)) (see sequence in Supp Table 4). Actin promoter sequences were amplified from *H. exemplaris* genomic DNA, *He*TDR1 cDNA from RNA of *H. exemplaris* adult tardigrades and mCherry from a mCherry containing plasmid. *He*-Act-*He*TDR1-GFP and Act5C-mCherry plasmids (2 μg/μL in milliQ water each) were co-injected in 20 starved *H. exemplaris* adults maintained in an in-house made PDMS injection chamber using Quartz micropipets. After 1h of microinjection, animals are let to recover in Volvic mineral water for 15min to 1h15. In order to get the plasmid into cells, tardigrades are next electroporated using NEPA21 Super Electroporator (Nepa Gene). Electric shock was carried out in 0.7X Optimem (Gibco, Thermofisher Life Sciences) with settings from (S. Tanaka, Aoki, and Arakawa 2023 [\[4\]](#)). Hoechst 33342 20 μM for 2 days or 40 μM for 1 day was also added to mineral water (Volvic, France) for live staining of the nucleus. Animals were immobilized using carbonated water and imaged by confocal microscopy (Zeiss (LSM 880 and AiryScan module) with x40 and x63 lens) using Zenblack software.

Expression of TDR1-GFP fusion proteins in human U2OS cells

Expression plasmids for fusion proteins of GFP and tardigrade proteins were constructed by Gibson assembly into pEGFP-N1 (Clontech) of the tardigrade cDNA (obtained by gene synthesis from Integrated DNA) or ordered from TwistBiosciences. Full nucleotide sequences of fusion proteins are provided as supplementary information (Supp Table 4). Plasmids were transfected into human U2OS cells (ATCC HTB-96) by Amaxa electroporation with Nucleofector™ Kit V (Lonza) and plated in 6 well plates containing glass slides.

Immunolabeling of phospho-H2AX foci in response to Bleomycin treatment and image analysis

Two days after transfection, Bleomycin sulfate-treated (treatment was for 1h with 1 μg/mL Bleomycin sulfate) or control cells were rinsed three times with PBS and fixed with 3.7% formaldehyde in PBS for 15 min at room temperature, rinsed three times with PBS, permeabilized with PBS, 0.5% Triton for 15 min, blocked with PBS, 0.1% Tween, 5% Fetal Calf Serum and incubated for 1h30 with specific anti-GFP (1 in 200 dilution of GFP Chicken polyclonal. #ab13970, Abcam) and anti-phospho H2AX (1 in 800 dilution of BW301, Merck) antibodies. After three PBS,

0.1% Tween washes, cells were incubated with secondary anti-chicken (Alexa Fluor® 488 Donkey Anti-Chicken. Reference: 703-546-155, Jackson ImmunoResearch) and anti-mouse (CyTM3 Goat Anti-Mouse. Reference: 115-167-003, Jackson ImmunoResearch) antibodies. After three PBS, 0.1% Tween washes, cells were incubated with Hoechst solution (11534886, Invitrogen) diluted 1/5000 in PBS, 0.1% Tween and mounted with ProLongTM Glass Antifade Mountant (P36982, Invitrogen). Cells were next imaged by confocal microscopy (Zeiss LSM 880) using Zenblack software and x40 lens in AiryScan mode acquisition of 7x7 contiguous XY fields and a Z-stack of 30 images at 0.1 µm intervals. Z-stacks were maximum projected and analyzed with Zen Blue software (v2.3) to automatically segment nuclei (using Hoechst staining), identify GFP-positive nuclei and count phospho-H2AX foci within each nucleus. When phospho-H2AX staining occupied more than a 1/3 of the nucleus surface, the number of foci was arbitrarily fixed as >400. Statistical significance of the difference in numbers of phospho-H2AX foci was measured with the non-parametric, rank-based Kruskal-Wallis test using GraphPad Prism (v9.3.1).

SEM of *P. fairbanksi* adults and eggs

Adults and eggs specimens were fixed with 2.5% glutaraldehyde in Volvic mineral water for 1h and washed three times with distilled water. The adults were put in microporous capsules and the eggs were filtered on Isopore membrane filters. The samples were dehydrated in ethanol series (50%, 70%, 90% and 100%). Then critical point (Leica CPD300, PTME MNHN) was used to dry them. Adults and membranes with eggs were deposited on carbon adhesive on the SEM stubs, coated with platinum (Leica EM ACE600 coater PTME MNHN) and examined using a scanning electron microscope (Hitachi SU3500, PTME MNHN).

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Data availability

All DNA sequencing data reported in the manuscript have been deposited in the NCBI SRA database and are accessible with Bioproject ID PRJNA997229. The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE ([Perez-Riverol 2022](https://www.ebi.ac.uk/pride/)) partner repository with the dataset identifier PXD043897.

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Reviewer #1 (Public Review):

Summary:

The manuscript aims to provide insights into the mediators and mechanisms underlying tardigrade radiation tolerance. The authors start by assessing the effect of ionizing radiation (IR) on the tardigrade lab species, *H. exemplaris*, as well as the ability of this organism to recover from this stress - specifically they look at DNA double and single strand breaks. They go on to characterize the response of *H. exemplaris* and two other tardigrade species to IR at the transcriptomic level. Excitingly, the authors identify a novel gene/protein called TDR1 (tardigrade DNA damage response protein 1). They carefully assess the induction of expression/enrichment of this gene/protein using a combination of transcriptomics and biochemistry - even going so far as to use a translational inhibitor to confirm the de novo production of this protein. TDR1 binds DNA in vitro and co-localizes with DNA in tardigrades.

Reverse genetics in tardigrades is difficult, thus the authors use a heterologous system (human cells) to express TDR1 in. They find that when transiently expressed TDR1 helps improve human cell resistance to IR.

This work is a masterclass in integrative biology incorporating a holistic set of approaches spanning next-gen sequencing, organismal biology, biochemistry, and cell biology. I think the importance of the findings is suitable and honestly, I find very little to critique in their experimental approaches.

Overall, I find this to be one of the more compelling papers on tardigrade stress-tolerance I have read.

<https://doi.org/10.7554/eLife.92621.2.sa2>

Reviewer #3 (Public Review):

Summary:

This paper describes transcriptomes from three tardigrade species with or without treatment with ionizing radiation (IR). The authors show that IR produces numerous single strand and double strand breaks as expected and that these are substantially repaired within 4-8 hours. Treatment with IR induces strong upregulation of transcripts from numerous DNA repair proteins, and from the newly described protein TDR1 with homologs in both Hypsibiodea and Macrobiotodea superfamilies. The authors show that TDR1 transcription produces newly translated TDR1 protein, which can bind DNA and co-localizes with DNA in the nucleus. At higher concentrations TDR appears to form aggregates with DNA, which might be relevant to a possible function in DNA damage repair. When introduced into human U2OS cells treated with the radiomimetic drug bleomycin, TDR1 reduces the number of double-strand breaks as

detected by gamma H2AX spots. This paper will be of interest to the DNA repair field and to radiobiologists.

Strengths:

The paper is well-written and provides solid evidence of the upregulation of DNA repair enzymes after irradiation of tardigrades, as well as upregulation of the TRD1 protein. The reduction of gamma-H2A.X spots in U2OS cells after expression of TRD1 supports a role in a DNA damage.

Weaknesses:

Genetic tools are still being developed in tardigrades, so there is no mutant phenotype to support a DNA repair function for TRD1, but this may be available soon.

<https://doi.org/10.7554/eLife.92621.2.sa1>

Reviewer #4 (Public Review):

In this study, Anoud et al. show convincing results of genes involved in the radio-resistance of tardigrades. With transcriptomics, they found many genes involved in DNA repair pathways to be overexpressed after ionizing radiation. In addition, they found RNF146 coding for a ubiquitin ligase, and genes of the AMNP family. Finally, they more deeply characterized one upregulated gene that they named TDR1 (Tardigrade DNA damage Response 1) which seems specific to tardigrades. With proteomics they verified these results. They show that TDR1 binds DNA in vitro and co-localize with DNA in tardigrades. Because of the difficulties of carrying reverse genetics in tardigrades, the authors showed in vitro that human cells expressing TDR1 led to a reduced number of phospho-H2AX foci (indicating DNA damages) when treated with Bleomycin. Based on these results, the authors suggested that TDR1 interacts with DNA and might regulate chromosomal organization and favors DNA repair.

Strengths:

The paper provides solid evidence of the upregulation of DNA repair enzymes after irradiation of tardigrades, as well as upregulation of the TRD1 protein.

The reduction of gamma-H2A.X spots in U2OS cells after expression of TRD1 supports a role in a DNA damage.

The shown interaction of TDR1 with DNA.

Weaknesses:

No reverse genetics to support a DNA repair function for TRD1, even if I recognize that these remain difficult to carry in tardigrades.

No pulse field electrophoresis gels to show DNA damages in tardigrades, which remain apparently challenging to perform in tardigrades.

After revision, the manuscript gained in structure, and in precision.

Overall, the manuscript provides valuable and convincing results contributing to our knowledge of tardigrade radio resistance. While reverse genetics remain difficult to carry in tardigrades, the authors used the alternative approach to investigate TDR1 function in vitro in human cells.

This study illustrates integrative biology as it combines a set of different methodologies including next-generation sequencing, transcriptomic and proteomic analyses,

immunohistochemistry, immunolabelling, in vitro assays and SEM. According to me, the quality and importance of the results make it of interest to the fields of DNA repair, radiobiology, and radio resistance.

<https://doi.org/10.7554/eLife.92621.2.sa0>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

The manuscript "comparative transcriptomics reveal a novel tardigrade specific DNA binding protein induced in response to ionizing radiation" aims to provide insights into the mediators and mechanisms underlying tardigrade radiation tolerance. The authors start by assessing the effect of ionizing radiation (IR) on the tardigrade lab species, H. exemplaris, as well as the ability of this organism to recover from this stress - specifically, they look at DNA double and single-strand breaks. They go on to characterize the response of H. exemplaris and two other tardigrade species to IR at the transcriptomic level. Excitingly, the authors identify a novel gene/protein called TDR1 (tardigrade DNA damage response protein 1). They carefully assess the induction of expression/enrichment of this gene/protein using a combination of transcriptomics and biochemistry - even going so far as to use a translational inhibitor to confirm the de novo production of this protein. TDR1 binds DNA in vitro and co-localizes with DNA in tardigrades.

Reverse genetics in tardigrades is difficult, thus the authors use a heterologous system (human cells) to express TDR1 in. They find that when transiently expressed TDR1 helps improve human cell resistance to IR.

This work is a masterclass in integrative biology incorporating a holistic set of approaches spanning next-gen sequencing, organismal biology, biochemistry, and cell biology. I find very little to critique in their experimental approaches.

Strengths:

- (1) Use of trans/interdisciplinary approaches ('omics, molecular biology, biochemistry, organismal biology)*
- (2) Careful probing of TDR1 expression/enrichment*
- (3) Identification of a completely novel protein seemingly involved in tardigrade radio-tolerance.*
- (4) Use of multiple, diverse, tardigrade species of 'omics comparison.*

Weaknesses:

- (1) No reverse genetics in tardigrades - all insights into TDR1 function from heterologous cell culture system.*
- (2) Weak discussion of Dsup's role in preventing DNA damage in light of DNA damage levels measured in this manuscript.*

(3) Missing sequence data which is essential for making a complete review of the work.

Overall, I find this to be one of the more compelling papers on tardigrade stress-tolerance I have read. I believe there are points still that the authors should address, but I think the editor would do well to give the authors a chance to address these points as I find this manuscript highly insightful and novel.

We thank the reviewer for his comments.

We agree that it will be important to further investigate the role of Dsup in radio-tolerance. We briefly mentioned this point in the discussion (p14). Our findings show that tardigrades undergo DNA damage at levels roughly similar to radio-sensitive organisms and therefore support a major role for DNA repair in the maintenance of genome integrity after exposure to IR. Nevertheless, we believe that more precise quantification of DNA damage may still reveal a contribution of genome protection to radio-tolerance of tardigrades compared to radio-sensitive organisms. Dsup loss of function experiments in tardigrades would clearly be the best way to assess this possibility. In the absence of experiments directly addressing the function of Dsup, we prefer to refrain from drawing any firm conclusion on prevention of DNA damage by Dsup and thus to keep a more open position. In any case, as discussed in the text, we note that Dsup has only been reported in Hypsibioidae and other molecular players, such as TDR1, are likely involved in radio-tolerance in other tardigrade species.

The sequence data can be accessed at the NCBI SRA database with Bioproject ID PRJNA997229.

Reviewer #3 (Public Review):

Summary:

This paper describes transcriptomes from three tardigrade species with or without treatment with ionizing radiation (IR). The authors show that IR produces numerous single-strand and double-strand breaks as expected and that these are substantially repaired within 4-8 hours. Treatment with IR induces strong upregulation of transcripts from numerous DNA repair proteins including Dsup specific to the Hypsibioidae superfamily. Transcripts from the newly described protein TDR1 with homologs in both Hypsibioidae and Macrobiotidae superfamilies are also strongly upregulated. They show that TDR1 transcription produces newly translated TDR1 protein, which can bind DNA and co-localizes with DNA in the nucleus. At higher concentrations, TDR appears to form aggregates with DNA, which might be relevant to a possible function in DNA damage repair. When introduced into human U2OS cells treated with bleomycin, TDR1 reduces the number of double-strand breaks as detected by gamma H2A spots. This paper will be of interest to the DNA repair field and to radiobiologists.

Strengths:

The paper is well-written and provides solid evidence of the upregulation of DNA repair enzymes after irradiation of tardigrades, as well as upregulation of the TDR1 protein. The reduction of gamma-H2A.X spots in U2OS cells after expression of TDR1 supports a role in DNA damage.

Weaknesses:

Genetic tools are still being developed in tardigrades, so there is no mutant phenotype to support a DNA repair function for TDR1, but this may be available soon.

We thank the reviewer for his comments.

Reviewer #4 (Public Review):

The manuscript brings convincing results regarding genes involved in the radio-resistance of tardigrades. It is nicely written and the authors used different techniques to study these genes. There are sometimes problems with the structure of the manuscript but these could be easily solved. According to me, there are also some points which should be clarified in the result sections. The discussion section is clear but could be more detailed, although some results were actually discussed in the results section. I wish that the authors would go deeper in the comparison with other IR-resistant eucaryotes. Overall, this is a very nice study and of interest to researchers studying molecular mechanisms of ionizing radiation resistance.

I have two small suggestions regarding the content of the study itself.

(1) I think the study would benefit from the analyses of a gene tree (if feasible) in order to verify if TDR1 is indeed tardigrade-specific.

(2) It would be appreciated to indicate the expression level of the different genes discussed in the study, using, for example, transcript per millions (TPMs). Recommendations for the authors: please note that you control which revisions to undertake from the public reviews and recommendations for the authors

We thank the reviewer for his comments.

(1) To identify TDR1 homologous sequences in non-tardigrade species, we conducted extensive homology searches using multiple homology-based approaches (Blastp and Diamond against the NCBI non-redundant protein sequences (nr) database and hmmsearch against the EBI reference proteomes), which failed to identify TDR1 homologs in non-tardigrade ecdysozoans, thus strongly supporting that TDR1 is indeed tardigrade-specific.

To be clearer in the manuscript, we now state the absence of hits for TDR1 in non-tardigrade ecdysozoans. Given the absence of homologs in non-tardigrade species, it is not possible to make a gene tree with non-tardigrade species.

(2) To further document expression levels (which were already available from the Tables in the initial submission), we added MAplots (representing log2foldchange and logNormalized read counts) in the supplementary materials (Supp Figure 3 and Supp Figure 8). These additional figures clearly document that the DNA repair genes discussed in the main text and TDR1 are highly expressed genes after IR and after Bleomycin treatment.

We thank the reviewer for his comments.

(1) It has always seemed strange to me that tardigrades accumulate just as much DNA damage as any other organism when irradiated and yet their Dsup protein is supposed to shield and protect their DNA from damage. Perhaps this is an appropriate time for this idea to be reconsidered given the Dsup was NOT induced by IR in this study and the authors found that their animals incurred just as much damage as other biological systems. While Dsup is clearly not the focus of this manuscript, it is the protein most associated with tardigrade radio-tolerance and I would argue this new paper would call into question previous conclusions made about Dsup.

Recommendations for the authors:

We thank the reviewer for his comments.

(1) It has always seemed strange to me that tardigrades accumulate just as much DNA damage as any other organism when irradiated and yet their Dsup protein is supposed to shield and protect their DNA from damage. Perhaps this is an appropriate time for this idea to be reconsidered given the Dsup was NOT induced by IR in this study and the authors found that their animals incurred just as much damage as other biological systems. While Dsup is clearly not the focus of this manuscript, it is the protein most associated with tardigrade radio-tolerance and I would argue this new paper would call into question previous conclusions made about Dsup.

We agree that it will be important to further investigate the role of Dsup in radio-tolerance. We briefly mentioned this point in the discussion (p14). Our findings show that tardigrades undergo DNA damage at levels roughly similar to radio-sensitive organisms and therefore support a major role for DNA repair in the maintenance of genome integrity after exposure to IR. Nevertheless, we believe that more precise quantification of DNA damage may still reveal a contribution of genome protection to radio-tolerance of tardigrades compared to radio-sensitive organisms. Dsup loss of function experiments in tardigrades would clearly be the best way to assess this possibility. In the absence of experiments directly addressing the function of Dsup, we prefer to refrain from drawing any firm conclusion on prevention of DNA damage by Dsup and thus to keep a more open position. In any case, as discussed in the text, we note that Dsup has only been reported in Hypsibioidae and other molecular players, such as TDR1, are likely involved in radio-tolerance in other tardigrade species.

(2) While reverse genetics are difficult in tardigrades, they are not impossible, and RNAi can be used to good effect in these animals. In fact several authors on this manuscript have used RNAi to examine the necessity of genes in tardigrade stress tolerance in the past. Was an attempt made to RNAi TDR1? If not, why? With the large amount of work that the authors put into showing the sufficiency of TDR1 for increasing radiotolerance in cell culture, one would think looking at necessity in tardigrades would be of great interest. If RNAi was performed, what were the results? Even a negative result here is informative since a protein can be sufficient but not necessary for a function - if this were the case it would mean tardigrades have some redundant mechanism(s) for surviving radiation exposure beyond TDR1.

We have attempted RNAi experiments targeting TDR1 or a mix of DNA repair genes (including XRCC5) and examined response to a bleomycin treatment of 2 weeks. Unfortunately, we could not distinguish any difference between uninjected animals and animals injected with TDR1 dsRNAs, or the mix of DNA repair genes dsRNAs. We concluded that, bleomycin treatment, that we used because it is much easier to perform than irradiation, was perhaps not the best way to assay a potential impact of RNAi on survival since it required long term treatment for several days during which the effect of RNAi may have waned. Another attempt was therefore made injecting with TDR1 or control GFP dsRNAs and exposing animals to a 2000Gy IR treatment. We noticed that the viability was lower after injection with GFP dsRNAs than with TDR1 dsRNAs (likely due to problems we had with the injection needle during injections). The next day, animals were irradiated and we observed after 24h that animals injected with GFP dsRNAs exhibited higher lethality rates than animals injected with TDR1 dsRNAs or uninjected animals. We found that this set of experiments were not conclusive. Our current experimental set up will make it difficult to distinguish lethality due to injections from lethality due to potentially decreased resistance to

Reviewer #1 (Recommendations For The Authors):

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the efficiency of target gene knockdown, as it is very challenging given the low amount of biological material available and the poor expression of these genes without irradiation). From a practical point of view, performing these experiments is thus very challenging. We nevertheless agree that, in future work, further experimentation is needed to examine the impact of knock-down by RNAi of TDR1 or of other genes such as DNA repair genes or Dsup, in tardigrade DNA repair and survival after IR. Gene knock-out with CRISPR-Cas9 is a very promising alternative to RNAi given that studies in mutant lines will eliminate the confounding effect of lethality due to injections.

(3) Regarding the U2OS experiments. I have several questions/points of clarification:

a. Were survival/proliferation levels tested or only H2AX foci? I think that showing decreased H2AX foci (fewer double-stranded breaks) correlates with higher survival rates would be important.

In the experiments reported in Figure 6, cells were transiently transfected with expression vectors and we did not examine the impact on survival rates. U2OS cells are resistant to high doses of Bleomycin and testing survival would require longer exposure at much higher concentrations (Buscemi et al, 2014, PMID: 25486478). In order to try and better address an impact on cell survival, we therefore generated populations of cells stably expressing the candidate tardigrade proteins fused to GFP. Despite trying different experiment conditions for treatment with Bleomycin, we could not detect a reproducibly significant benefit on cell survival for any of the tardigrade proteins tested, including RvDsup which was used as a positive control (since it was previously reported to improve cell survival in response to X-rays). One possibility is that the analysis should be performed in clones and not in populations of cells with heterogeneous expression levels of the tardigrade protein tested. For example, expression levels of the tardigrade protein needed to reduce the number of phospho-H2AX foci in response to DNA damage may interfere with cell division. We note that in the original Dsup paper, the benefit of RvDsup on cell survival was reported in specific transgenic clones. Experiments in different biological systems have also started to document toxic effects of RvDsup expression, illustrating the challenge, when performing experiments in heterologous systems, to achieve suitable expression levels of the tested protein. Trying to perform such a finer analysis, in our opinion, would go beyond the scope of our manuscript and will be best addressed in future studies. We are therefore careful in the text not to make any claim on the benefit of TDR1 expression on cell survival in response to Bleomycin in human cultured cells.

(b) From the methods I am a bit confused as to how the images were treated/foci quantified. With the automatic segmentation and foci identification, is this done through the entire Z-series or a single layer? If the latter then I am not sure the results are meaningful, since we do not know how many foci might be present in other layers of the nuclei analyzed. If the former, please clarify this in the method since it is a very important consideration.

We have acquired images throughout the entire Z-series and edited the text to make it more clear ; We now write: “ Z-stacks were maximum projected and analyzed with Zen Blue software (v2.3)...”. To limit the time needed for image analysis, we have generated an artificial image by projecting the entire Z-series into a single image and counted foci in that single maximum projection image. Although there are potential drawbacks, such as potentially only counting one focus when two foci are superposed along the Z axis, this approach overcomes the limitations of quantification from a single layer. We further ensured statistical robustness of the analysis by performing quantification from several independent fields of the labelled cells and several independent biological replicates ($n \geq 3$ as now specified in the legend of figure 6a).

(c) RvDsup reduced levels of HXA1 foci in these experiments, however, HeDsup was not found to be enriched in the transcriptomic analysis performed here. Was there a reason HeDsup was not used in the cell-based experiments? One could argue that RvDsup is from a different species of tardigrade, but it is a bit concerning that an ortholog of a protein found NOT to be induced by radiation exposure seems to perform as well (if not better) than some versions of TDR1.

RvDsup is the protein initially shown to increase survival of human HEK293 cells treated with X-rays and reduce the number of phospho-H2AX foci induced: it was therefore used as a positive control in our experiments. The sequence of HeDsup is only poorly similar to RvDsup (with 26% identity) and activity of HeDsup in cultured cells has not been reported before. We therefore believe that HeDsup is not well suited to provide a positive control for the experiments performed in our manuscript.

(d) From the methods, it seems that cells were treated with Bleomycin and then immediately fixed without any sort of recovery time. In this short timeframe, the presence of TDR1 appears to be enough to deal with a substantial amount of double-stranded breaks (as evidenced by the reduced number of HXA1 foci). Does this make sense? How quickly could one expect DNA repair machinery to make significant progress in resolving damaged DNA? This response seems much faster than what was observed in tardigrades. Perhaps the authors to comment on this.

Kinetic studies in human cells show extremely rapid repair of DNA double-strand breaks. Sensing of DNA double strand breaks by PARP proteins takes place within seconds after irradiation by IR (Pandey and Black, 2021, PMID: 33674152). NHEJ is then observed to take place by formation of 53BP1 foci within 15 minutes (Schultz et al, 2000, PMID: 11134068). The number of phospho-H2AX and 53BP1 foci peaks at 30 minutes and starts declining thereafter, showing that at a significant number of sites, DNA repair is proceeding very rapidly (by NHEJ). Although we are not aware of any studies of DNA repair kinetics in U2OS cells after addition of Bleomycin, DNA damage must be instantaneous and further take place during exposure to the drug in parallel to DNA repair, which would be expected to have similar kinetics than after irradiation with IR.

In our experiments, several mechanisms may be involved in reducing the number of phospho-H2AX foci induced by Bleomycin, such as DNA protection (for Dsup expression) or stimulation of DNA repair (for RNF146 expression). For TDR1, the molecular mechanism involved remains to be determined. Given our finding that TDR1 can form aggregates with DNA, an additional possibility is that clustering of phospho-H2AX foci is induced.

(4) I could not find the sequences of the TDR1 proteins studied here. I did find the cDNA sequence of HeTDR1 in the final supplementary file, but not the other TDR1 orthologs. In the place where it appeared the TDR1 sequences from other tardigrades should be there were very short segments of the HETDR1 sequence. All sequences of proteins used in this study should be easily accessible to the reader and reviewers as it is not possible to review this work without accessing the sequences.

Our apologies for the inappropriate documentation of TDR1 sequences in the original manuscript. As requested, we have now included the TDR1 sequences in the Supplementary Table 4.

(5) Likewise, the RNA sequence data is said to be deposited in NCBI under PRJNA997229, but I do not find this available on NCBI.

The RNA sequence data was deposited in NCBI under the indicated reference before submission of the manuscript. The data has now been released and is fully available on NCBI.

(6) A few typographical errors: e.g., Page 10 - sentence 4 has two periods "." or page 14 which has an open parenthesis that is not closed.

These typos have been corrected in the revised manuscript.

Reviewer #3 (Recommendations For The Authors):

We thank the reviewer for his comments.

In Figure 4C, what fraction of the 50 genes upregulated in all species and treatments are DNA repair genes? Is there any other notable commonality between these 50 genes? The bulk of upregulated genes are specific to a species and to treatment with IR or bleomycin. What fraction of DNA repair genes are specific to a species or treatment?

The results in Figure 4C on the 50 putative orthologous genes upregulated in all species and treatments are further detailed in supp Figure 10. The legend to supp Figure 10 now provides the requested information: 14/50 genes are DNA repair genes and the other notable commonality is that 21/50 are "stress response genes". We did not further breakdown the analysis to evaluate the fraction of DNA repair genes specific to a species or treatment. It will be interesting to gather data in more species to shed light on the evolutionary history of DNA repair gene regulation in response to IR.

How does the suite of upregulated tardigrade DNA repair proteins after IR or bleomycin compare with DNA or repair proteins upregulated under similar treatments in human cells? Are they quantitatively or qualitatively different, or both?

There is a great wealth of studies documenting genes differentially expressed in human cells in response to IR (e.g. Borrás-Fresneda et al, 2016, PMID: 27245205; Rieger and Chu, 2004, PMID: 15356296; Budwoeth et al, 2012, PMID: 23144912 ; Rashi-Elkeles et al, 2011, PMID: 21795128; Jen and Cheung, 2003, PMID: 12915489...). Upregulation of DNA repair and cell cycle genes is commonly found. However, the number of DNA repair genes induced is always very limited and fold stimulation very modest compared to the massive upregulation observed in tardigrades.

On page 14, please explain the acronym BER. Do the authors mean Base Excision Repair? Or something else?

As assumed by the reviewer, the acronym BER stands for Base Excision Repair. The acronym has been removed from the main text and replaced by the full name.

We thank the reviewer for his comments.

Abstract:

The abstract is fine. What was hard to grasp at the beginning is why TDR1 gene was named that way. It should be clearer that this study decided to further focus on that gene, one of the most overexpressed gene after IR, with an unknown function. Then maybe introduce that it was found to be unique to tardigrade and to interact with DNA. Therefore, it was named TDR1.

Reviewer #4 (Recommendations For The Authors):

We thank the reviewer for his comments.

Abstract:

The abstract is fine. What was hard to grasp at the beginning is why TDR1 gene was named that way. It should be clearer that this study decided to further focus on that gene, one of the most overexpressed gene after IR, with an unknown function. Then maybe introduce that it was found to be unique to tardigrade and to interact with DNA. Therefore, it was named TDR1.

Introduction:

The introduction has been modified according to the suggestions of Reviewer#4 below. One of the suggested references, Nicolas et al 2023 from the Van Doninck lab, was published while our manuscript was under review and cannot be considered as background information for our study.

1st paragraph:

The study is on tardigrades, I found it strange that the first paragraph is on D. radiodurans. I think it is fine to mention what is known in bacteria and eucaryotes but we should already know what will be the main topic in the first paragraph of the introduction. Some details about D. radiodurans seem less important and distracting from the main topic (3D conformation).

2nd paragraph:

When mentioning radio-resistant eucaryotes the authors do not mention the larvae of the anhydrobiotic insect Polypedilum vanderplanki. Stating that the mechanisms of resistance are poorly characterized should perhaps be nuanced. There are some recent studies on D. radiodurans (Ujaoney et al., 2017) the insect P. vanderplanki (Ryabova et al., 2017), tardigrades (Kamilari et al., 2019), and rotifers (Nicolas et al., 2023, Moris et al., 2023). Perhaps these papers are worth indicating that if mechanisms are not elucidated yet, recent studies suggest some actors involved in their resistance. Regarding the sentence stating that DNA repair rather than DNA protection plays a predominant role in the radio-resistance of bdelloid rotifers should also be nuanced. Indeed, many chaperones, antioxidants were mentioned to play a role in the radio-resistance of bdelloid rotifers (Moris et al., 2023). The authors mentioned the reference Hespeels et al., 2023 which is not found in their list of references, I am not sure which paper they refer to. The last sentence of the second paragraph does not mean much. I am not sure what the authors want to state with this. Perhaps they should specify if they mean that the function of many other genes overexpressed after IR remains unknown.

Still, in the second paragraph, the authors focus on rotifers. They also do not mention what is known in the insect P. vanderplanki, which should be added. They still do not mention tardigrades. I think it is nice to first start with eucaryotes and then focus on tardigrades but as I mentioned before it would help to understand the aim of the paper if the first paragraph mentioned briefly the tardigrades and then could go into detail in the third paragraph.

3rd paragraph:

The sentence starting "with over 1400 species" best to remove from it "but they can differ in their resistance" and start the next sentence with that.

4th paragraph:

Very clear, we finally understand what is the focus of the manuscript.

5th paragraph:

*Very clear. The authors should mention the names of the three studied species. Here, *A. antarcticus* is missing. The sentence "Further analyses in *H. exemplaris*... showed that TDR1 protein is present and upregulated". The authors should mention in which conditions the protein is upregulated. In that paragraph the authors mention phospho-H2AX: it might be good to introduce its functions before in the introduction (it is mentioned in the second sentence of the results: best to move it to the introduction).*

Results:

There are a few sentences in this section which rather discuss the results than describe them. I think the manuscript might gain in quality if these interpretations of the results are moved into the discussion section. That would make the result section more concise and the discussion enriched.

For instance, I suggest to move these sentences into the discussion:

- *"the finding of persistent DSBs in gonads at 72h.... likely explains..."*
- *"suggesting that (i) DNA synthesis..."*
- *" Phospho-H2AX....also suggested"*
- *"Moreover, expression of TDR1-GFP..., supporting the potential role of TDR1 proteins..."*
- *"our results suggest that RNF146 upregulation could contribute..."*
- *"AMNP gene g12777 was shown to increase...Based on our results, it is possible that..."*

Interpretations mentioned here above were always introduced cautiously (-"suggesting that (i) DNA synthesis..." ; -" Phospho-H2AX....also suggested" ; -"Moreover, expression of TDR1-GFP..., supporting the potential role of TDR1 proteins..." ; -"our results suggest that RNF146 upregulation could contribute..."). These cautious interpretations were usually important in deciding next steps of the work. We therefore believe it is important to mention these interpretations in the results section to clearly expose the milestones marking the progression of the study.

For some results, they were directly discussed in the results section for the sake of concision (for example -"the finding of persistent DSBs in gonads at 72h.... likely explains..." ; -"AMNP gene g12777 was shown to increase...Based on our results, it is possible that...") since, in our opinion, there was no need to mention them again in the main discussion.

Some other parts could be good to be moved into the introduction:

- *"Previous studies have indicated that irradiation with IR increases expression of Rad51,..." none of the actors involved in DNA repair are mentioned in the introduction. Also, change resistant into resistance*

- *"A. antarcticus ... known for its resistant to high doses of UV..."*

We have moved these parts to the introduction as recommended.

It was in O. areolatus.... that the first demonstration..."

This piece of information is somewhat anecdotal. We choose to keep it here in the results section. This information on the radio-resistance of the species *P. areolatus* is only relevant at this specific step of the study because it encouraged us to consider that *P. fairbanksi*, which we isolated fortuitously, would be a good model species for studying radio-resistance of tardigrades.

Here are some additional comments/suggestions on the result section:

1st section

- *Remove the Gross et al., 2018 from the sentence "using confocal microscopy", it looks otherwise that these results are from their study, not yours.*

We have changed the text to make it clear that this is indeed a finding of Gross et al which was previously made in non-irradiated tardigrades. We replicated this finding, which showed that the protocol was working appropriately, and that we could use this control result for comparison with irradiated animals. We apologize for this confusion.

The text now states: "Using confocal microscopy, we could detect DNA synthesis in replicating intestinal cells of control animals, as previously shown by (Gross et al. 2018)."

2nd section

- *It is confusing what has been found induced by IR and/or by Bleomycin.*
- *I think it might help if the authors first present what is induced after IR, then write if it is similar after Bleomycin. Especially since they start to do it in the first paragraph of that section. However, they only mention TDR1 in the second paragraph dedicated to Bleomycin treatment which is confusing as it is also overexpressed after IR. It is also not clear if RNF146 is also induced by Bleomycin.*

As recommended, the text presents first what is induced after IR and then what is induced by Bleomycin in the following paragraph. When reporting results with Bleomycin, we have provided a global assessment of what is common to both treatments in Supp Figure 3 and in Supp Table 3. In this figure, we also specifically highlighted several key genes of DNA repair induced by both treatments. These are also mentioned in the text (p8) to illustrate the point that many key DNA repair genes are common to both treatments. We have now added RNF146 to that list as recommended.

- *Regarding TDR1, it is not clear when introduced in the text as "promising candidate" why it is the case. It is clear in the figures but perhaps the authors should explain why they chose these genes for further analyses: high log2foldchange and expression level for instance. Regarding that last comment, it would be interesting to have an idea about the expression level of the genes with high log2foldchange. In Figures 2, 3, and 4 the pvalue and log2foldchange are represented but not the expression level (ideally Transcript per Millions).*

These values would give an additional idea on the importance of that gene. While looking at the figures, it is unclear why you did not further characterize other genes with high log2foldchange (some with even hints of their function): the mentioned RNF146, macroH2A1 (not even mentioned in the results), some genes unannotated in the figures with likely unknown functions,

When selecting genes of interest, we did indeed take into account high expression levels. To more clearly document expression levels (which were already available from the Tables), we added MAplots (representing log2foldchange and logNormalized read counts) in the supplementary materials (Supp Figure 3 and Supp Figure 8).

- It is also unclear at that stage why you named it "Tardigrade DNA damage response protein", as it is characterized as DNA repair/damage proteins by specific GO id or is it based on your downstream analyses, I think it might be worth to quickly mention the reason of that name.*

The name illustrates two points which were already characteristic at this point in time of the study i.e. 1) it is a tardigrade specific protein and 2) it is induced in response to DNA damage.

- Regarding the BLAST analyses the protein was searched in C. elegans, D. melanogaster and H. sapiens. Why only these three species? What were the threshold values used for these analyses. As mentioned in the main comment, it would be worth searching species phylogenetically close to tardigrades to verify if it is well-tardigrade specific. Did you try to make a gene tree, after looking for a conserved domain (using hmmersearch)?*

As indicated in the methods section, the "Tardigrade-specific" annotation was determined by absence of hits after high-throughput alignment (with diamond using –ultrasensitive-option) on the NCBI nr database and absence of hits after blast search on C. elegans, D. melanogaster and H. sapiens proteomes as a complementary criterion (the latter blast search was primarily performed to enrich for functional annotations). Based on these criteria, TDR1 was annotated as "Tardigrade-specific". As stated in the text, we also searched for TDR1 related sequences with 1) blastp (which is more sensitive than diamond) on the NCBI nr database and 2) HMMER on Reference Proteomes, and no hits were found among non-tardigrade ecdysozoans organisms, confirming TDR1 is specific to tardigrades. For Blast search for example, there were five hits in non-ecdysozoans organisms (two cephalochordates, one mollusc and two echinoderma). The blastp and HMMER results are now included in the revised supplementary material (Supp Table 5). These very few hits in species phylogenetically distant from tardigrades cannot be taken to support the existence of TDR1 genes outside tardigrades.

To be clearer in the manuscript, we now state the absence of hits for TDR1 in non-tardigrade ecdysozoans. Given the absence of homologs in non-tardigrade species, it is not possible to make a gene tree with non-tardigrade species.

- Page 9: "Proteins extracts from H. exemplaris... at 4h and 24h..." I think this sentence can be removed as this is mentioned again 2 paragraphs after: "...we conducted an unbiased proteome analysis... at 4h..." The log2foldchange threshold mentioned for the proteomic analyses is 0.3: why this threshold, was it chosen randomly?*

This threshold is commonly used when considering log2foldchange with the technology used in our study, an isobaric multiplexed quantitative proteomic strategy which is known to

compress ratios (Hogrebe et al. 2018).

- *Page 10:*

It would be good for more clarity to indicate at the beginning of the new section which species were investigated after IR or Bleomycin treatment.

TDR1 homologs in the other tardigrade species were identified based on what? Best reciprocal hit?

As indicated in the methods section of the manuscript, we searched for homologs in other tardigrade species by BLAST. A best reciprocal hit approach was not performed to try to determine which homologs might be orthologs. In particular, most TDR1 homologs identified are known from transcriptome assemblies and high-contiguity genome assemblies are needed to more confidently identify orthology (using synteny). The results of the BLASTP search are now provided as supplementary material (Supp Table 5).

Preliminary experiments indicated that A. antarcticus and P. fairbanksi survived exposure to 1000 Gy: is there a supplementary graph showing this?

We have corrected the text to avoid any confusion. We have not rigorously examined the dose-dependent survival of *P. fairbanksi* in response to irradiation. Text was changed to: “We found by visual inspection of animals after IR that *A. antarcticus* and *P. fairbanksi* readily survived exposure to 1000 Gy.”

- *Page 11:*

"A set of 50 genes was upregulated in the three species": please be precise if only after IR.

Done

These genes cannot be the same as they are from different species. Did the author mean that they are coding for similar proteins? It might be good to give some more details even if the supplementary figure is mentioned.

Obviously, these genes are putative orthologs. We have changed the text to:

” a set of 50 putative orthologous genes was upregulated in response to IR in all three species”

Discussion:

- *General comment: the discussion is focused mainly on TDR1, it would be nice to also discuss the other results: DNA repair genes, RNF146.*

A whole paragraph is devoted to discussion of results on DNA repair genes and RNF146. We have extended that discussion following on the suggestion of the reviewer. In particular, we have explicitly mentioned the apparent paradox that XRCC5 and XRCC6, which are among the most highly stimulated genes at the mRNA level, only display modest upregulation at the protein level. Although further studies would be needed to examine the mechanisms involved, we propose that upregulation of RNF146, whose human homolog has been shown to drive degradation of PARylated XRCC5 and XRCC6 proteins in response to IR (Kang et al. 2011), may be responsible for higher degradation rates and may thus counterbalance increased levels of protein synthesis.

- *Pulse field electrophoresis would be nice to be performed. It has been used to assess DSBs in bdelloid rotifers, is it possible in tardigrades?*

As stated in the discussion, we believe that it would be challenging to perform pulse field electrophoresis in tardigrades. However, if possible, these experiments would certainly bring invaluable information to complement our analysis of DNA damage induced by IR.

- *"By comparative transcriptomics": please rephrase that sentence.*
- *Proteins acting early in DNA repair: I am not sure I understand this sentence. Actors as ligases act not at the beginning of the repair pathways.*

Well noted. We have removed ligases from the list.

- *It is confusing that the authors mention NHEJ and double-strand break repair pathways as different pathways. There are 2 main pathways to repair DSBs: NHEJ and HR. It would be nice to add a reference to the sentence "PARP proteins act as sensors of DNA damage etc."*

A typo in the sentence gave rise to the misleading suggestion that NHEJ is not a double strand repair pathway. It has been corrected.

A reference has been added for PARP proteins.

- *It would be nice if the authors can explain deeper their suggestion that degradation of DNA repair actors is essential for tardigrade IR resistance.*

We have expanded this part of the discussion and hope that it is clearer.

“For XRCC5 and XRCC6, our study established, by two independent methods, proteomics and Western blot analyses, that the stimulation at the protein level could be much more modest (6 and 20-fold at most (Supp Figure 6) than at the RNA level (420 and 90 fold respectively). This finding suggests that the abundance of DNA repair proteins does not simply increase massively to quantitatively match high numbers of DNA damages. Interestingly, in response to IR, the RNF146 ubiquitin ligase was also found to be strongly upregulated. RNF146 was previously shown to interact with PARylated XRCC5 and XRCC6 and to target them for degradation by the ubiquitin-proteasome system (Kang et al. 2011). To explain the lower fold stimulation of XRCC5 and XRCC6 at the protein levels, it is therefore tempting to speculate that, XRCC5 and XRCC6 protein levels (and perhaps that of other scaffolding complexes of DNA repair as well) are regulated by a dynamic balance of synthesis, promoted by gene overexpression, and degradation, made possible by RNF146 upregulation. Consistent with this hypothesis, we found that, similar to human RNF146 (Kang et al. 2011), He-RNF146 expression in human cells reduced the number of phospho-H2AX foci detected in response to Bleomycin (Figure 6).”

- *Page 15: Please add a reference for the sentence "Functional analysis of promotor sequences in transgenic tardigrades etc."*

The reference has been added to fix this omission.

Material and Methods:

Small comments:

- 40 μm mesh: space missing
- 100 μm mesh: space missing
- (for Bleomycin)): parenthesis missing
- remove "as indicated in the text"
- The investigated time points after radiation need to be clearly stated in the method section. It is also unclear in the IR and Bleomycin section which tardigrades were treated with what. Not all were treated with Bleomycin.

The small comments above have been fixed in the revised version of the manuscript.

- Page 21: please precise the coverage of the RNA sequencing

Statistics on mapping of RNAseq reads are now provided in Supp Table 10.

- Page 22: Was any read trimming performed? Anything about the quality check of the reads?

Trimming was conducted using trimmomatic (v0.39) and quality check using FastQC (v. ?) This information has been added to the Methods section.

- Were the analyses confirmed by a second approach: for instance, EdgeR? Deseq2 and EdgeR do not always have the same results. For more robust analyses it is advised to use both.

Differential transcriptome analyses were conducted with DESeq2 only. The robustness of our identification of differentially expressed genes in response to IR stems from performing comparative analyses in three different species, rather than from using two bioinformatics pipelines in a single species. We also note that benchmarking reported in the initial DESeq2 paper showed that identification of differentially expressed genes with large log fold changes (which, as reported in our manuscript, is characteristic of many DNA repair genes in response to IR) is very consistent between DESeq2 and EdgeR.

Figures:

- Figure 2: Legend vertical dotted line does not indicate log2foldchange value of 4 in all panels: it would be good to indicate for panels a and c as well.

Figure 2 has been improved following on the suggestions of the reviewer. Dotted lines now show log2foldchange value of 2 in all panels (ie Fold Change of 4 as mentioned in the main text).

- Figure 2C: There are a few points with high log2foldchange which are not annotated: was it because nothing was found in the blast research? If yes, it would be good to indicate their functions. If not, it would be good to mention in

the discussion that there are some genes with still unknown functions which might play an important role in the resistance of tardigrades to IR.

The few points which are not annotated in figure 2c can now be found in Supp Table 3 Some of them have no hit in Blast search, some others such as BV898_09662 or BV898_07145 have hits on DNA repair genes as RBBP8/CtIP or XRCC6 respectively but are not annotated as such by eggNOG in KEGG pathway.

- *Figure 4C: Why not have included the response of P. fairbanksi to bleomycin? I guess it was not done, but it is unclear in the results and methods sections.*

P.fairbanksi response to bleomycin wasn't assessed as we didn't get enough animals to run the study. The method section has been modified to precise this point.